Sinais e Sistemas para Engenharia

MÓDULO 4 - ANÁLISE NO TEMPO DE SISTEMAS DISCRETOS

VERSÃO OCTAVE

PROFESSOR LUCIANO NEVES DA FONSECA

Um Sistema Discreto é normalmente descrito por uma equação de recorrência / algoritmo

Por exemplo, sejam os sistemas descritos pela seguintes equações:

Sistema A) A saída é igual a média entre a entrada atual, a entrada passada e a penúltima entrada

$$x[n] \xRightarrow{H_A(z)} y\,[n] = \frac{(x[n] + x[n-1] + x[n-2])}{3}$$

Sistema B) A saída é a soma de quatro vezes a entrada atual , mais três vezes a entrada anterior, mais duas vezes a penúltima, mais a antepenúltima entrada"

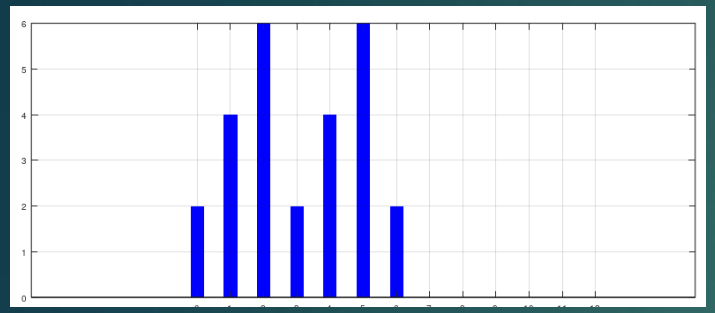
$$x[n] \xRightarrow{H_B(z)} y[n] = \frac{4x[n] + 3x[n-1] + 2x[n-2] + x[n-3]}{10}$$

Como calcular a resposta destes sistemas a diversos sinais de entrada x(n)?

Cálculo da resposta do Sistema A (a diversos sinais) através da equação de recorrência

$$x[n] \xRightarrow{H_A(z)} y[n] = \frac{(x[n] + x[n-1] + x[n-2])}{3}$$

```
>> x=[2 4 6 2 4 6 2 zeros(1,6)];
>> n=[0:length(x)-1];
>> bar(n,x,0.4,'b','edgecolor','none')
```



```
function y_out=rec1A(x_in)
    xi=[0 0 x_in];
    for (i=3:length(xi));
        yi(i)=(xi(i)+xi(i-1)+xi(i-2))/3;
    end
    y_out=yi(3:length(xi));
endfunction
```

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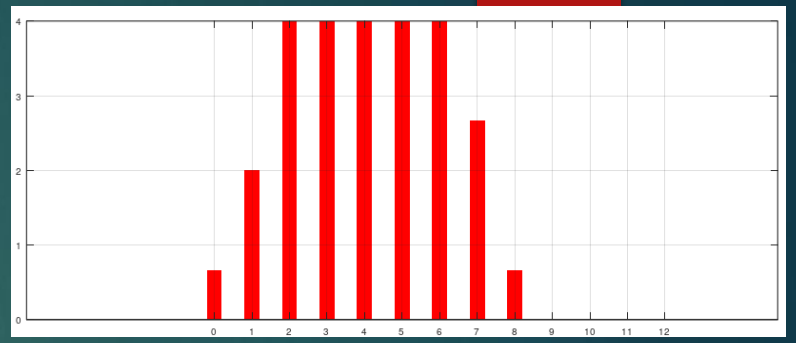

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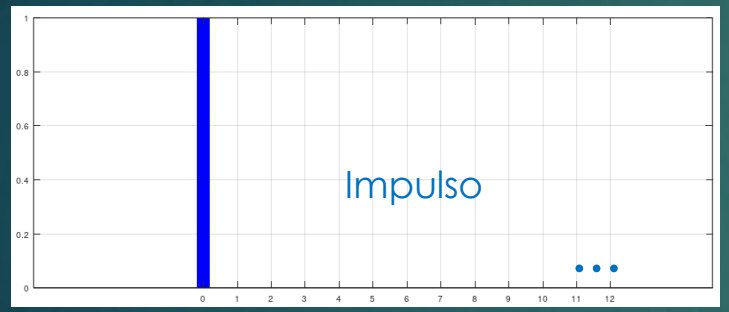


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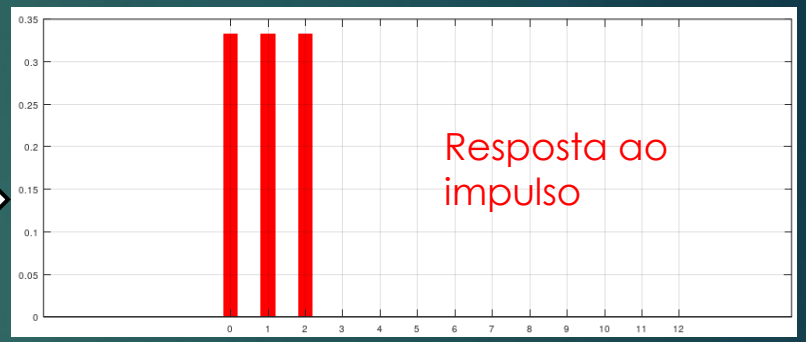


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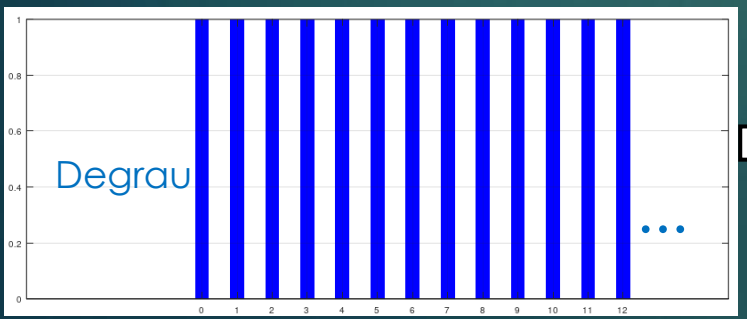


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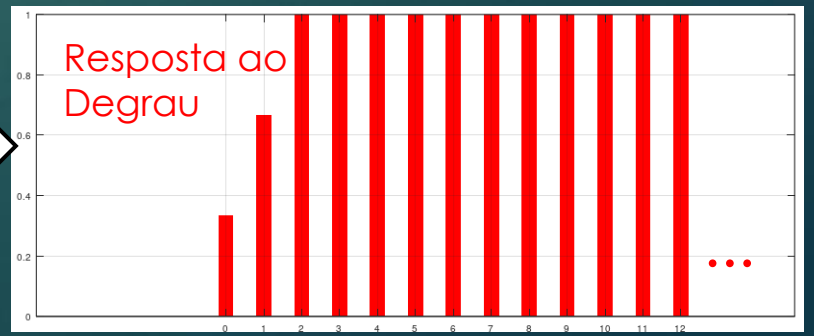

Cálculo da resposta do Sistema B (a diversos sinais) através da equação de recorrência

$$x[n] \xrightarrow{H_B(z)} y[n] = \frac{4x[n] + 3x[n-1] + 2x[n-2] + x[n-3]}{10}$$

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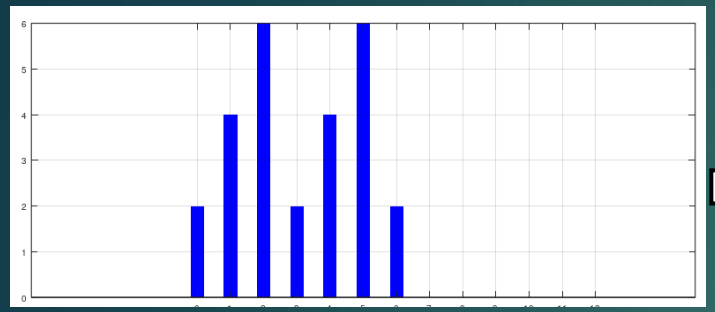

```
function y_out=rec1B(x_in)
    xi=[0 0 0 x_in];
    for (i=4:length(xi));
        yi(i)=(4*xi(i)+3*xi(i-1)+2*xi(i-2)+xi(i-3))/10;
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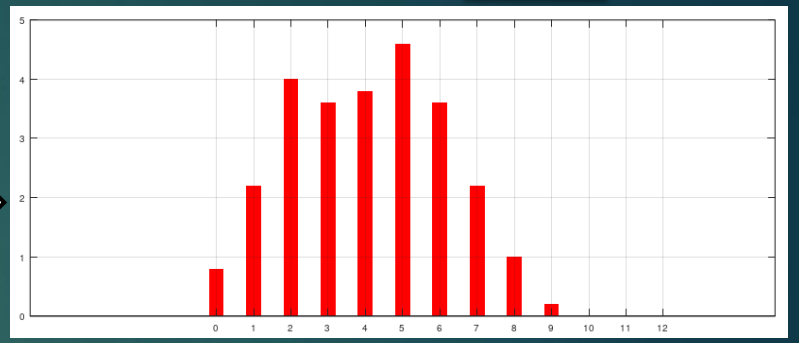

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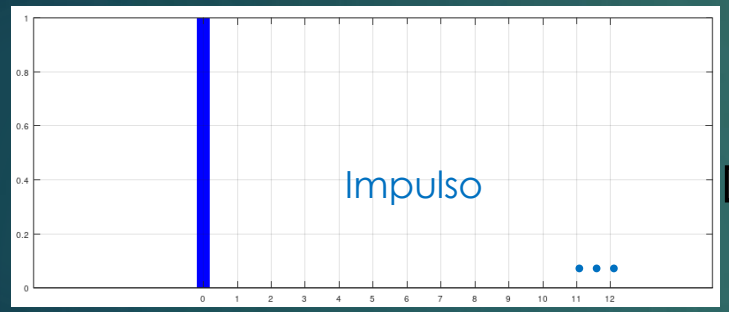


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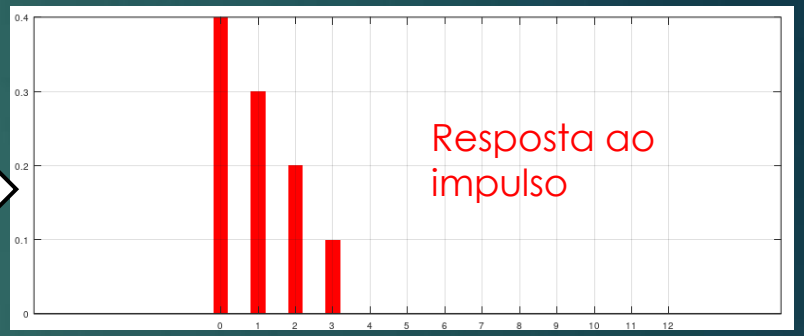


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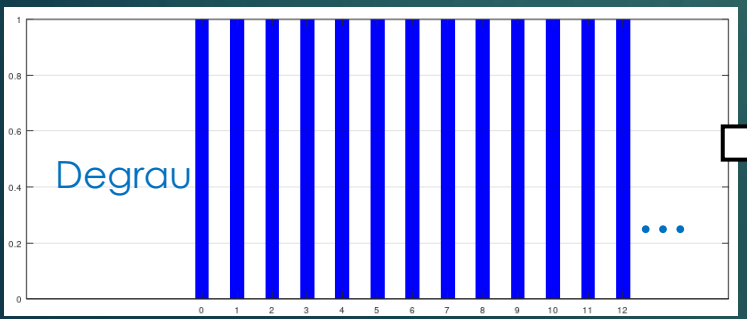


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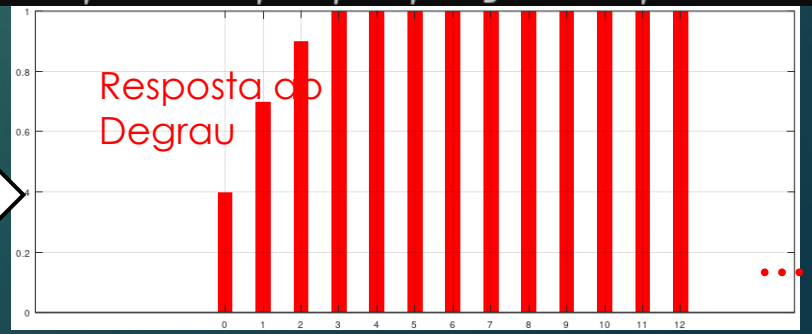


Representação de qualquer sinal a partir da soma de impulsos.

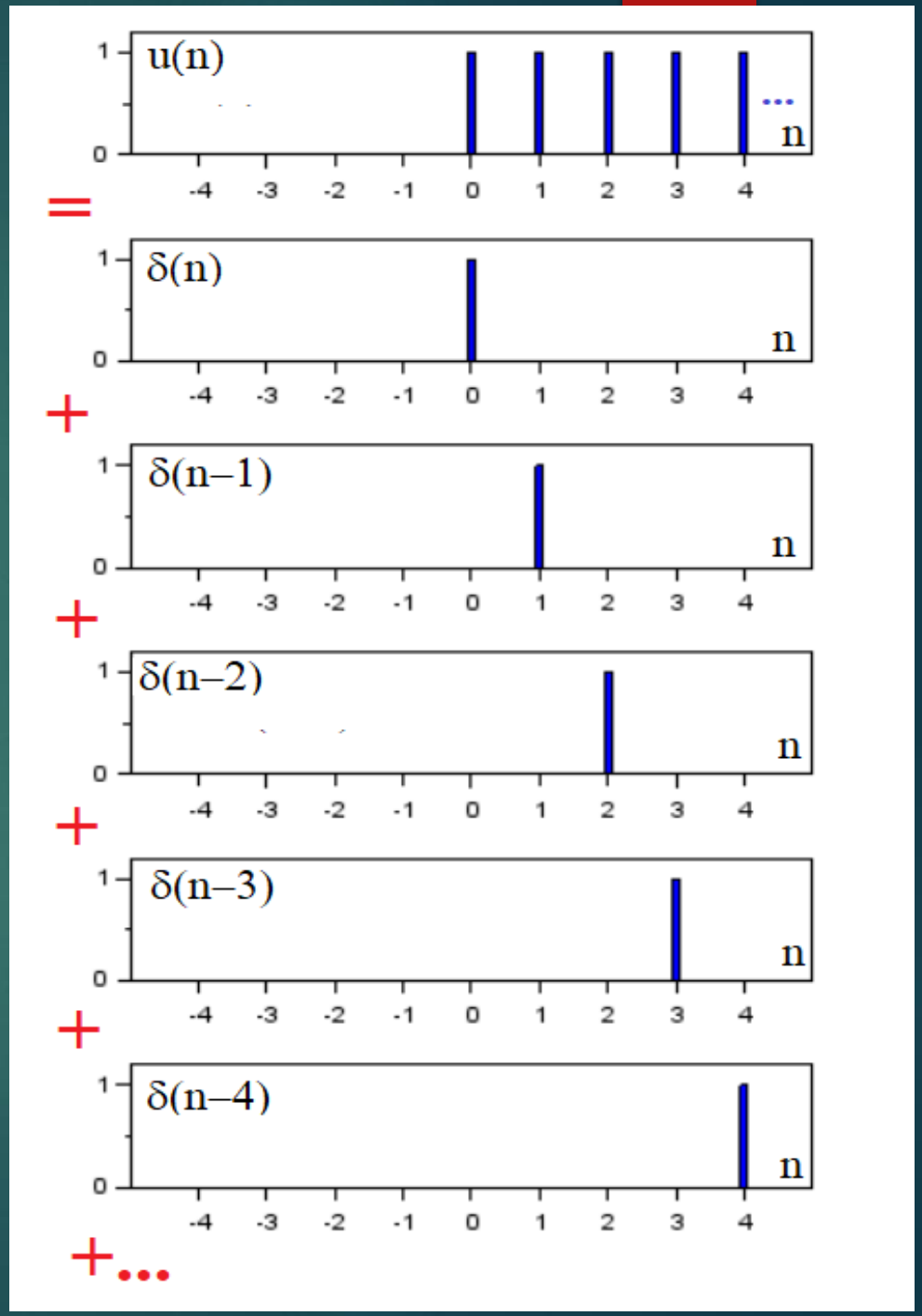
- O Degrau unitário $u(n)$ pode ser expresso com um soma de impulsos deslocados:

Degrau Discreto $u[n] = \begin{cases} 0; n < 0 \\ 1; n \geq 0 \end{cases}$

Impulso Discreto $\delta[n] \begin{cases} 1;\ n = 0 \\ 0;\ n \neq 0 \end{cases}$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-4] + \ldots$$



- Da mesma forma, o sinal $x(n)$ mostrado abaixo pode ser expresso como um soma de impulsos com amplitude variável, deslocados no tempo:

$$x[n] = [0, 0, 0, 0, 4, 1, 3, 7, 2] \quad (-4)$$

$$x[n] = \sum_{k=-4}^{4} x[k]\,\delta[n-k)]$$

$$x[n] = 4\delta[n] + \delta[n-1] + 3\delta[n-2] + 7\delta[n-3] + 2\delta[n-4].$$

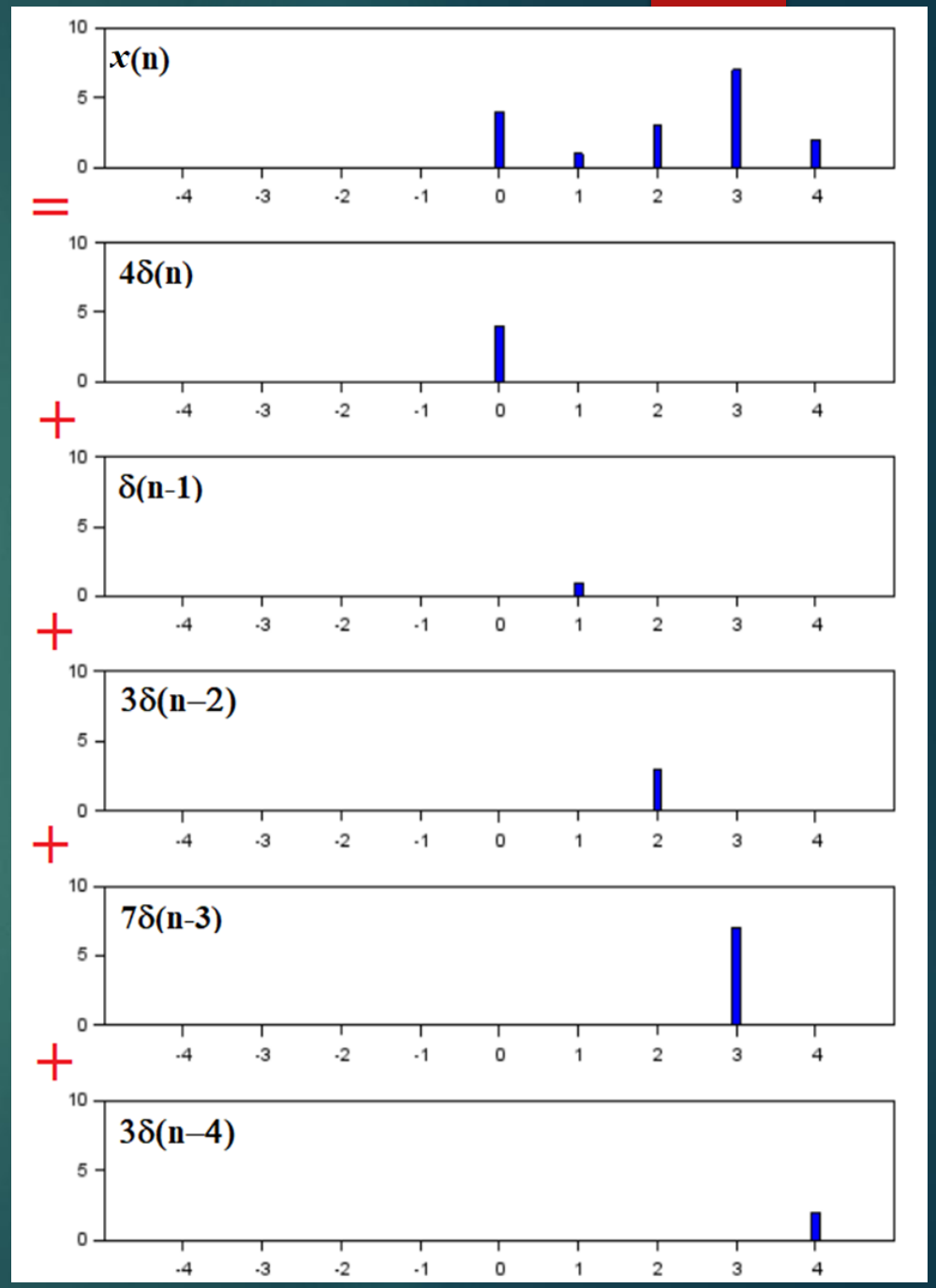
- De uma maneira geral , um sinal $x(n)$ qualquer pode ser expresso como um soma de impulsos com amplitude variável, deslocados no tempo:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Exemplo:

$$x[n] = 2n + 1; \quad n \geq 0$$

$$x[n] = \delta[n] + 3\delta[n-1] + 5\delta[n-2] + 7\delta[n-3] + 9\delta[\mathrm{n}-4] + \cdots$$



Resposta de um sistema LTI a uma soma ponderada de impulsos

Suponha um sistema discreto linear invariante no tempo, com função de transferência H(z). Se na entrada colocarmos o impulso contínuo δ(t), então a saída será h(n) (resposta ao impulso do sistema discreto)

Se o Sistema for linear invariante no tempo, o fato de conhecermos h(n) (a resposta ao impulso), nos permite calcular a resposta do sistema a qualquer outra entrada..

Como $\delta[n] \xRightarrow{H(z)\,.} h[n]$, sendo h(n) a resposta ao impulso

e $a\delta[n] \xRightarrow{H(z)} ah\,[n]$, já que a equação é linear

e $\delta[n-k] \xRightarrow{H(z)} h[n-k]$, já que equação é invariante no tempo

então $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$, já que x(n) pode ser escrito como uma soma de impulsos atrasados com amplitude x(k).

Então

$$\boxed{x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[\,n-k] \xRightarrow[\textit{Equação de Recorrência}]{H(z)} y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]}$$

A resposta de um sistema LTI a uma soma ponderada de impulsos será uma soma ponderada de respostas ao impulso!!

Convolução Discreta

$$x[n] \longrightarrow \boxed{\begin{matrix} \boldsymbol{Hz} \\ \boldsymbol{Eq.Recorrência} \end{matrix}} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Se conhecermos a resposta ao impulso δ(n), conheceremos a resposta do sistema a qualquer entrada x(n), através da soma ponderada das respostas ao impulso do sistema.

Esta soma é conhecida como Convolução Discreta.

$$y[n] = x[n] * h[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Assim, a resposta de um sistema a uma entrada x(n) é simplesmente a Convolução do sinal x[n] com a resposta do sistema ao impulso h[n]

A resposta de um sistema linear e invariante no tempo a uma entrada x[n] é a soma ponderada das respostas ao impulso deslocadas no tempo.

Notar que se $x(n)$ tiver N_x elementos e se $h(n)$ tiver N_h elementos, então $y(n)$ terá $N_x + N_h - 1$ Elementos

Calculo das respostas do Sistema A através da convolução com $h[n] = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$

$$x[n] \xrightarrow{H_A(z)} y[n] = \frac{(x[n] + x[n-1] + x[n-2])}{3}$$

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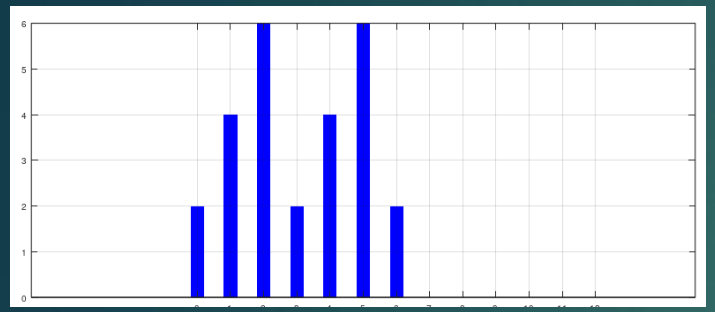

$H_A(z)$

```
>> h=[1/3 1/3 1/3];
```

```
>> y=conv(x,h);
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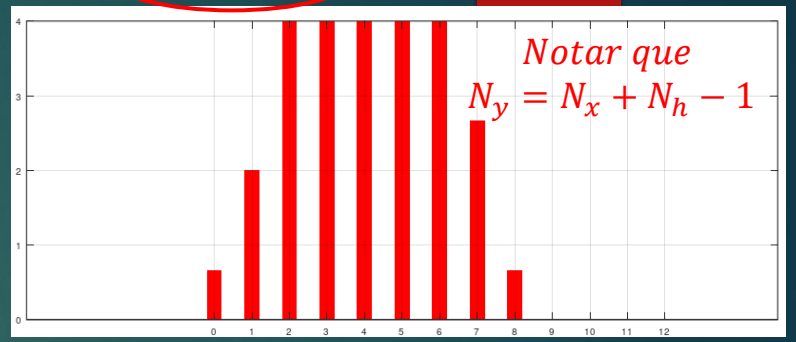

Notar que ny = nx +nh -1
(2 elementos a mais)
Estes 2 elementos podem ser ignorados

```
>> x=[1 zeros(1,12)];
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```

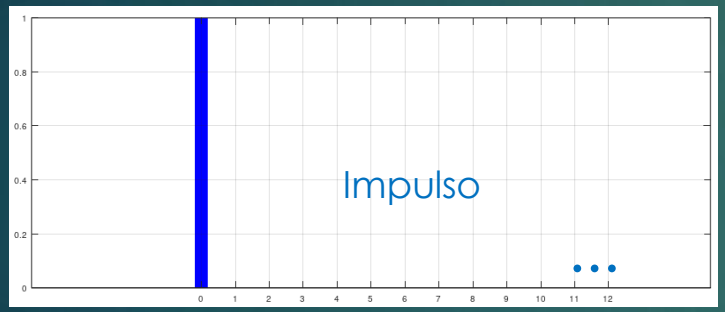

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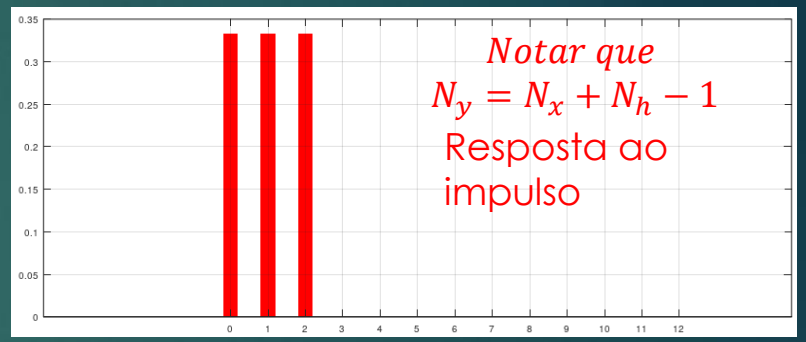


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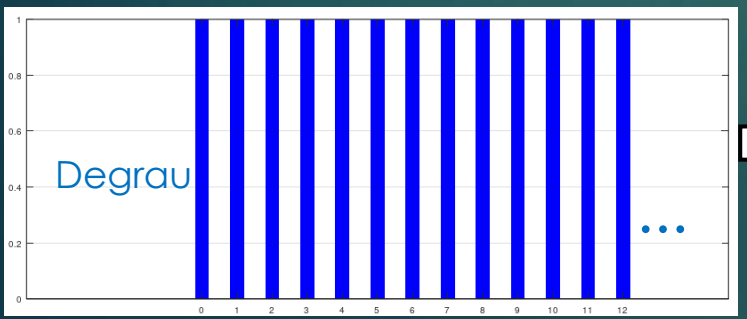

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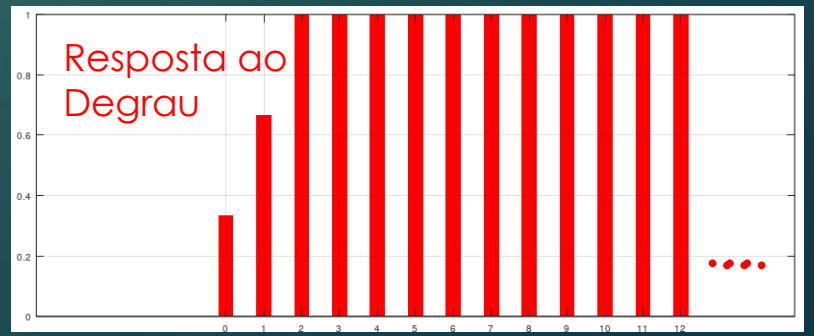

Calculo das respostas do Sistema B através da convolução com $h(n) = [3,2,1]$

$$x[n] \xrightarrow{H_B(z)} y[n] = \frac{4x[n] + 3x[n-1] + 2x[n-2] + x[n-3]}{10}$$

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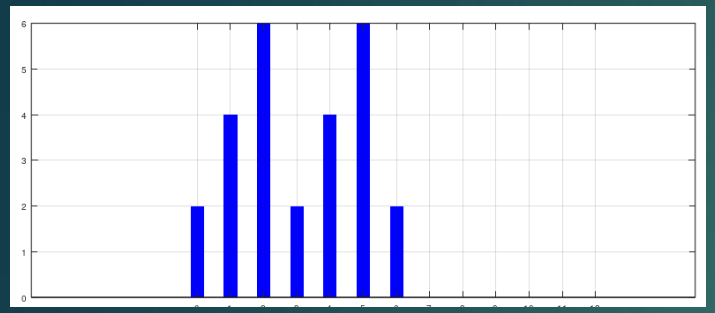

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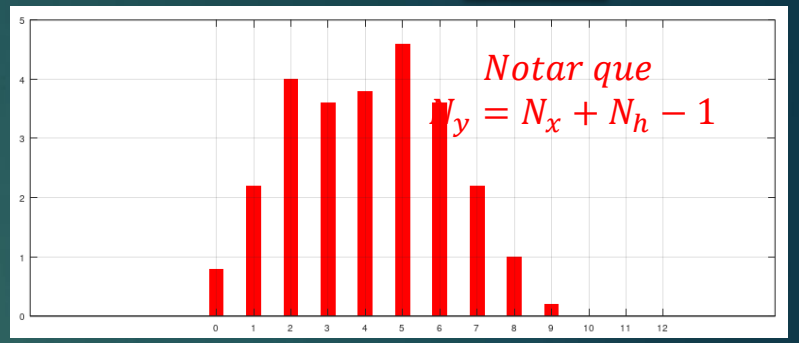


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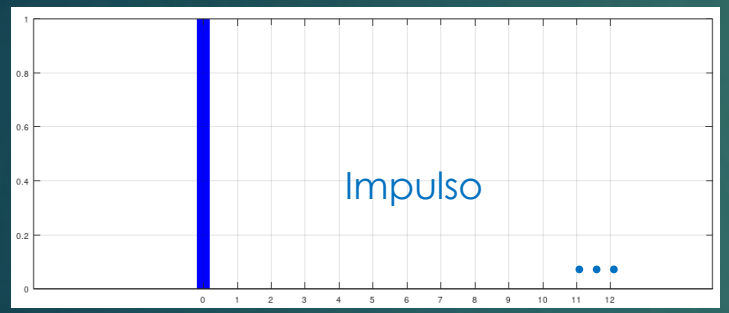

Notar que ny = nx +nh -1
(2 elementos a mais)
Vamos ignorar estes 2 elementos extras

$H_B(z)$

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>> h=[4/10 3/10 2/10 1/10];
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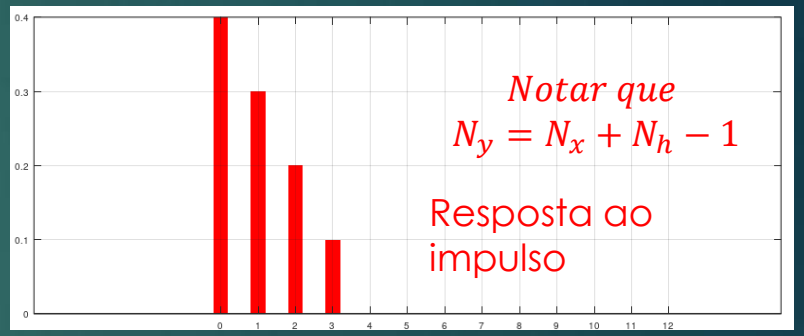


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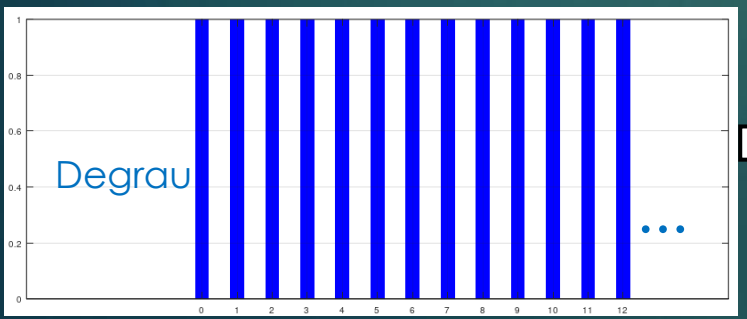

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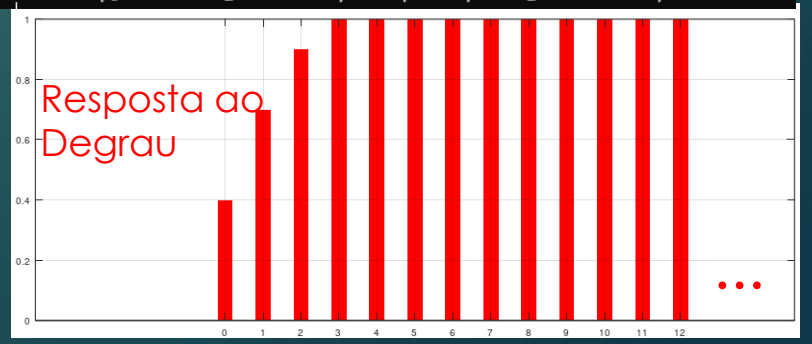


Tabela para o Cálculo Manual da Convolução Discreta

Calcular $y[n] = x[n] * h[n]$

$$\sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$x[n] = [1, -2, 3, -1]$ (0)

$h[n] = [3,2,1]$ (0) $\qquad h[-k] = [1,2,3]$ (-2)

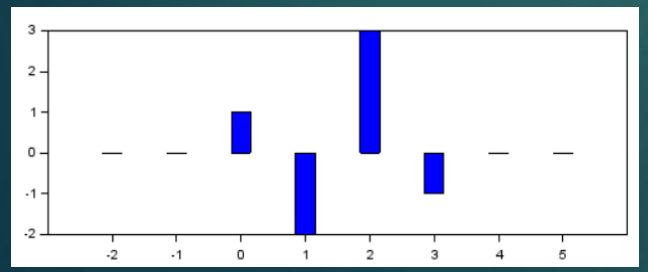
$y[n] = x[n] * h[n] = [3, -4, 6, 1, 1, -1]$ (0)

- 1) Inverter $h[k] \rightarrow h[-k]$
- 2) Deslocar $h[-k] \rightarrow h[n-k]$
- 3) Multiplicar $h[n-k]$ por $x[k]$ para cada k
- 4) Somar as parcela para todo n

-2	-1	0	1	2	3	4	5	y(n)	
		1	-2	3	-1				x(n)
1	2	3						3	0
	1	2	3					-4	1
		1	2	3				6	2
			1	2	3			1	3
				1	2	3		1	4
					1	2	3	-1	5

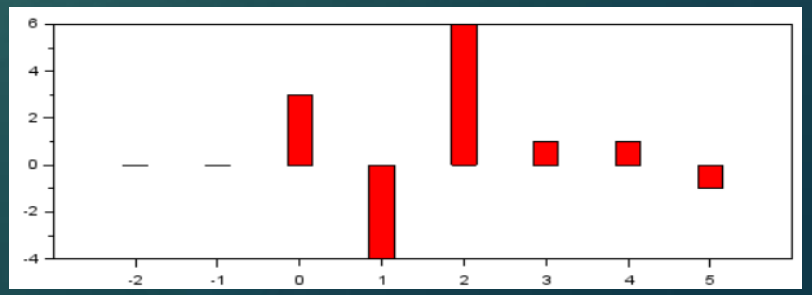
```
>> y=conv_fga(x,0,h,false)
y =

   3  -4   6   1   1  -1
```

```
>> x=[1,-2,3,-1];
```



$H_B(z)$

```
>> h=[3,2,1];
```



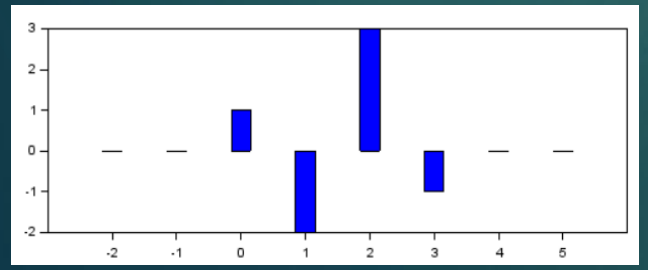
```
function [y_out,n_out]=conv_fga(x,xi,h,prt)
    nx = length(x);
    nh = length(h);
    i_ini = xi - nh +1;
    i_fin = xi + nx + nh - 2;
    ncols = i_fin-i_ini+1;
    nlins = nx + nh -1;;
    indice=[i_ini:i_fin];
    x_ext=zeros(1,ncols);
    for coluna=1:nx
        x_ext(coluna+nh-1)=x(coluna);
    end
    h_r = flip(h);
    h_des=zeros(nlins,ncols);
    for linha=1:nlins
        for i = 1:nh
            h_des(linha,i+linha-1)=h_r(i);
        end
       y_out(linha)=sum(x_ext.*h_des(linha,:));
    end
    for linha=1:nlins
       h_des(linha,ncols+1)=y_out(linha);
       h_des(linha,ncols+2)=xi+linha-1;
    end
    if (prt)
        disp(indice);
        disp(x_ext);
        disp(h_des);
    end
    n_out = h_des(:,ncols+2)';
endfunction
```

```
>> x=[1,-2,3,-1];
>> h=[3,2,1];
>> [y,n]=conv_fga(x,0,h,false)
y =

   3  -4   6   1   1  -1

n =

   0   1   2   3   4   5
```

```
>> [y,n]=conv_fga(x,0,h,true);
  -2  -1   0   1   2   3   4   5
   0   0   1  -2   3  -1   0   0
   1   2   3   0   0   0   0   0   3   0
   0   1   2   3   0   0   0   0  -4   1
   0   0   1   2   3   0   0   0   6   2
   0   0   0   1   2   3   0   0   1   3
   0   0   0   0   1   2   3   0   1   4
   0   0   0   0   0   1   2   3  -1   5
```

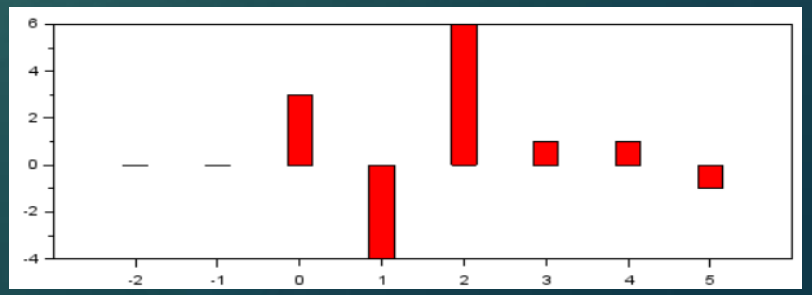
```
>> y=conv_fga(x,0,h,false)
y =

   3  -4   6   1   1  -1
```

```
>> x=[1,-2,3,-1];
```



$H_B(z)$

```
>> h=[3,2,1];
```



Sistemas discretos podem então ser descritos:

1) Resposta ao impulso $h[n]$ / Equação de Recorrência / Função de Transferência em Z

Resposta ao impulso $h[n]$

$h[n] = [4, -1, 5, 2, 3]$ (-1)

Equação de Recorrência

$y[n] = 4\,\mathrm{x}[n+1] - \mathrm{x}[n] + 5\mathrm{x}[n-1] + 2\mathrm{x}[n-2] + 3\mathrm{x}[n-3]$

Função de Transferência em Z

$y[n] = 4\,\mathrm{x}[n+1] - \mathrm{x}[n] + 5\mathrm{x}[n-1] + 2\mathrm{x}[n-2] + 3\mathrm{x}[n-3]$

$$Y(z) = 4z^1\,X(z) - X(z) + 5z^{-1}X(z) + 2z^{-2}X(z) + 3z^{-3}X(z)$$

$$Y(z) = \left(\mathbf{4z - 1 + 5z^{-1} + 2z^{-2} + 3z^{-3}}\right)X(z)$$

$$\boldsymbol{H(z) = \frac{Y(z)}{X(z)} = 4z - 1 + 5z^{-1} + 2z^{-2} + 3z^{-3}}$$

$x[n] \rightarrow \boldsymbol{X(z)}$

$x[n-1] \rightarrow \boldsymbol{z^{-1}X(z)}$

$x[n-2] \rightarrow \boldsymbol{z^{-2}X(z)}$

$x[n+1] \rightarrow \mathrm{z}\,\boldsymbol{X(z)}$

$y[n] \rightarrow \boldsymbol{Y(z)}$

$$\boldsymbol{H(z) = \frac{Y(z)}{X(z)}}$$

Sistemas discretos podem então ser descritos:

2) Resposta ao impulso h(n) / Equação de Recorrência / Função de Transferência em Z

Função de Transferência em Z

$$\boldsymbol{H(z) = 1 + 7z^{-1} + 3z^{-2}}$$

Equação de Recorrência

$$\boldsymbol{H(z) = \frac{Y(z)}{X(z)} = 1 + 7z^{-1} + 3z^{-2}}$$

$$Y(z) = \boldsymbol{X(z) + 7z^{-1}X(z) + 3z^{-2}}X(z)$$

$$y[n] = \mathrm{x}[n] + 7\mathrm{x}[n-1] + 3\mathrm{x}[n-2]$$

Resposta ao impulso h(n)

$$h[n] = [1,7,3]\ (0)$$

$$y[n] = \mathrm{x}[n] + 7\mathrm{x}[n-1] + 3\mathrm{x}[n-2]$$

$$x[n] \quad \rightarrow \boldsymbol{X(z)}$$

$$x[n-1] \rightarrow \boldsymbol{z^{-1}X(z)}$$

$$x[n-2] \rightarrow \boldsymbol{z^{-2}X(z)}$$

$$\boldsymbol{H(z) = \frac{Y(z)}{X(z)}}$$

Sistemas discretos podem então ser descritos:

3) Resposta ao impulso $h[n]$ / Equação de Recorrência / Função de Transferência em Z

Equação de Recorrência

$y[n] = 4\mathrm{x}[n-1] - \mathrm{x}[\mathrm{n}-2] + 5\mathrm{x}[\mathrm{n}-3]$

$x[n] \rightarrow \boldsymbol{X(z)}$

$x[n-1] \rightarrow \boldsymbol{z^{-1}X(z)}$

$x[n-2] \rightarrow \boldsymbol{z^{-2}X(z)}$

$x[n-3] \rightarrow \boldsymbol{z^{-3}X(z)}$

$$\boldsymbol{H(z) = \frac{Y(z)}{X(z)}}$$

Resposta ao impulso h(n)

$h[n] = [4, -1{,}5]\ (1) = [0, 4, -1{,}5]$

Função de Transferência em Z

$y[n] = 4\mathrm{x}[n-1] - \mathrm{x}[\mathrm{n}-2] + 5\mathrm{x}[\mathrm{n}-3]$

$Y(z) = 4z^{-1}X(z) - z^{-2}X(z) + 5z^{-3}X(z)$

$Y(z) = (4z^{-1} - z^{-2} + 5z^{-3})X(z)$

$$H(z) = \frac{Y(z)}{X(z)} = 4z^{-1} - z^{-2} + 5z^{-3}$$

$$\boldsymbol{H(z) = 4z^{-1} - z^{-2} + 5z^{-3}}$$

Sistemas discretos podem então ser descritos:

4) Resposta ao impulso $h[n]$ / Equação de Recorrência / Função de Transferência em Z

Equação de Recorrência

$$y[n] = 5x[n] + 3x[n-2] - \frac{3}{4}y[n-1]$$

Função de Transferência em Z

$$y[\mathrm{n}] = 5x[\mathrm{n}] + 3x[\mathrm{n}-2] - \frac{3}{4}y[\mathrm{n}-1]$$

$$Y(z) = 5X(z) + 3z^{-2}X(z) - \frac{3}{4}z^{-1}Y(z)$$

$$Y(z) + \frac{3}{4}z^{-1}Y(z) = 5X(z) + 3z^{-2}X(z)$$

$$\left(1 + \frac{3}{4}z^{-1}\right)Y(z) = (5 + 3z^{-2})X(z)$$

$$\boldsymbol{H(z)} = \frac{\boldsymbol{Y(z)}}{\boldsymbol{X(z)}} = \frac{5 + 3z^{-2}}{1 + \frac{3}{4}z^{-1}}$$

Resposta ao impulso h(n)

$h[n] = ?$

$x[n] \rightarrow \boldsymbol{X(z)}$

$x[n-2] \rightarrow \boldsymbol{z^{-2}X(z)}$

$y[n] \rightarrow \boldsymbol{Y(z)}$

$y[n-1] \rightarrow \boldsymbol{z^{-1}Y(z)}$

$$\boldsymbol{H(z)} = \frac{\boldsymbol{Y(z)}}{\boldsymbol{X(z)}}$$

Observar que neste exemplo a equação de recorrência a função de transferência do sistema H(z) agora tem denominador, pois depende de saídas passadas (realimentação)

Cálculo da resposta ao impulso de um sistema com retroalimentação, através da sua equação de recorrência

$y[n] = 5x[n] + 10x[n-2] - \frac{3}{4}y[n-1]$ $\quad y[-1] = 0$ (condição inicial!!)

$y[n] = [5.\ \ -3.75\ \ 12.81\ \ -9.61\ \ 7.21\ \ -5.41\ \ 4.05\ \ -3.04\ \ 2.28\ \ -1.71\ \ 1.28 \ldots]$

$$Y(z) = 5X(z) + 10z^{-2}X(z) - \frac{3}{4}z^{-1}Y(z) \rightarrow \quad H(z) = \frac{5 + 10z^{-2}}{1 + \frac{3}{4}z^{-1}}$$

```
function y_out=rec2(x_in,y_ini)
    xi=[0 0 x_in];
    yi=[ nan  y_ini];
    for (i=3:length(xi))
        yi(i)=5*xi(i)+10*xi(i-2)-3/4*yi(i-1);
    end
    y_out=yi(3:length(xi));
endfunction
```

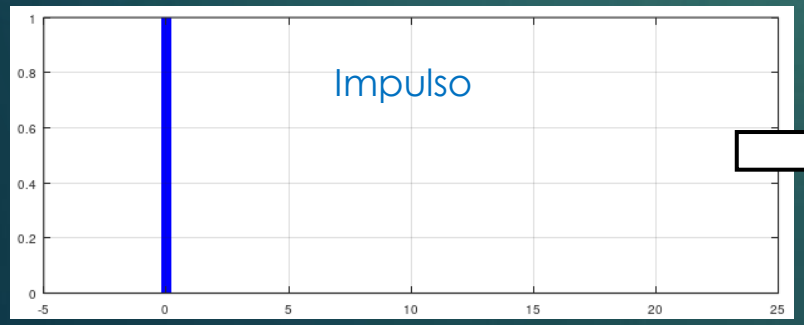
Notar que este sistema, por depender de saídas passadas (retroalimentação) , tem resposta ao impulso com duração infinita (IIR)

```
>> x=[1 zeros(1,20)];
>> n=[0:length(x)-1];
>> bar(n,x,0.4,'b','edgecolor','none')
```

```
>> n=[0:20];
>> x=f_d(n);
>> bar(n,x,0.4,'b','edgecolor','none')
```

$$H(z) = \frac{5 + 10z^{-2}}{1 + \frac{3}{4}z^{-1}}$$

```
>> bar(n,rec2(x,0),0.4,'r','edgecolor','none')
```

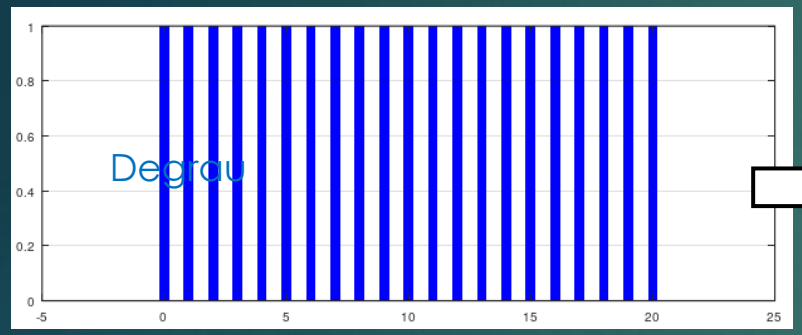



```
function y_out=rec2(x_in,y_ini)
    xi=[0 0 x_in];
    yi=[ nan  y_ini];
    for (i=3:length(xi))
        yi(i)=5*xi(i)+10*xi(i-2)-3/4*yi(i-1);
    end
    y_out=yi(3:length(xi));
endfunction
```

Exemplo: Cálculo da resposta de um sistema com retroalimentação, a uma entrada qualquer, através da equação de recorrência

$$y[n] = 5x[n] + 10x[n-1] - \frac{3}{4}y[n-1] \qquad y[-1] = 0$$

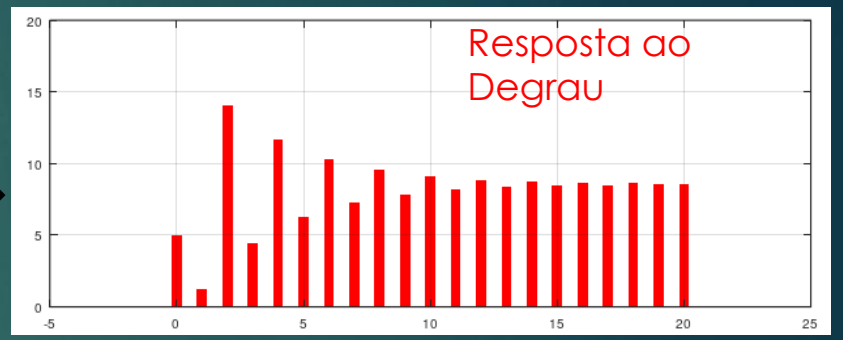
```
>> n=[0:20];
>> x=f_u(n);
>> bar(n,x,0.4,'b','edgecolor','none')
```



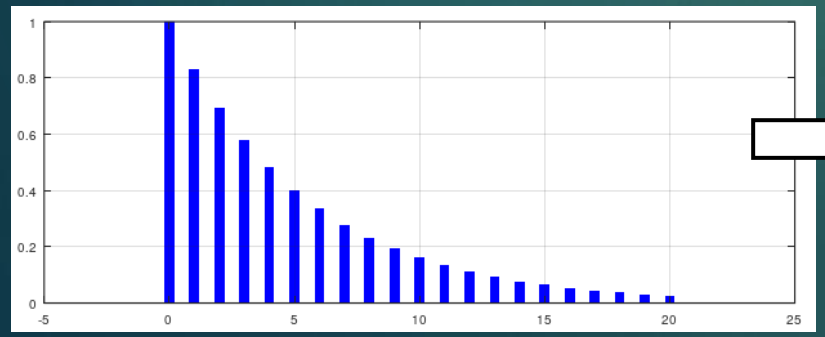
$$H(z) = \frac{5 + 10z^{-2}}{1 + \frac{3}{4}z^{-1}}$$

```
function y_out=rec2(x_in,y_ini)
    xi=[0 0 x_in];
    yi=[ nan  y_ini];
    for (i=3:length(xi))
        yi(i)=5*xi(i)+10*xi(i-2)-3/4*yi(i-1);
    end
    y_out=yi(3:length(xi));
endfunction
```

```
>> bar(n,rec2(x,0),0.4,'r','edgecolor','none')
```

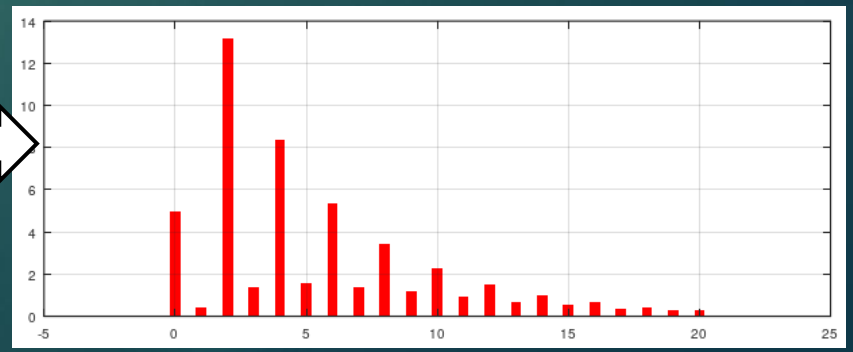



```
>> n=[0:20];
>> fx = @(t) (5/6).^t.*f_u(t);
>> x=fx(n);
>> bar(n,x,0.4,'b','edgecolor','none')
```



```
function y_out=rec2(x_in,y_ini)
    xi=[0 0 x_in];
    yi=[ nan  y_ini];
    for (i=3:length(xi))
        yi(i)=5*xi(i)+10*xi(i-2)-3/4*yi(i-1);
    end
    y_out=yi(3:length(xi));
endfunction
```

```
>> bar(n,rec2(x,0),0.4,'r','edgecolor','none')
```



Sistemas FIR e IIR

Sistemas FIR são sistemas diretos - "Finite Impulse Response"

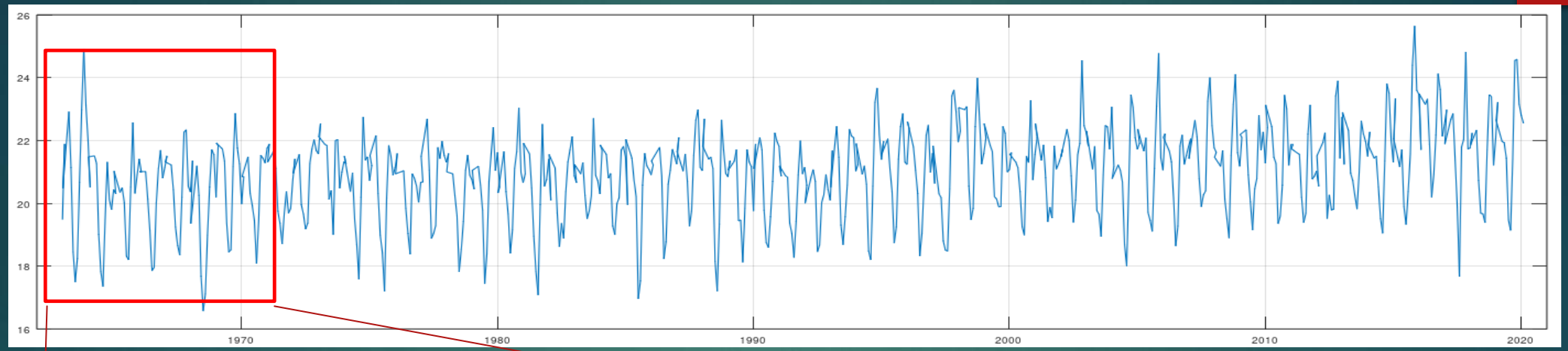
- São sistemas sem realimentação.
- A saída de sistemas FIR só depende das entradas (não depende das saídas passadas).
- A função de transferência um sistema FIR só tem numerador, não tem denominador

Sistemas IIR são sistemas realimentados "Infinite Impulse Response".

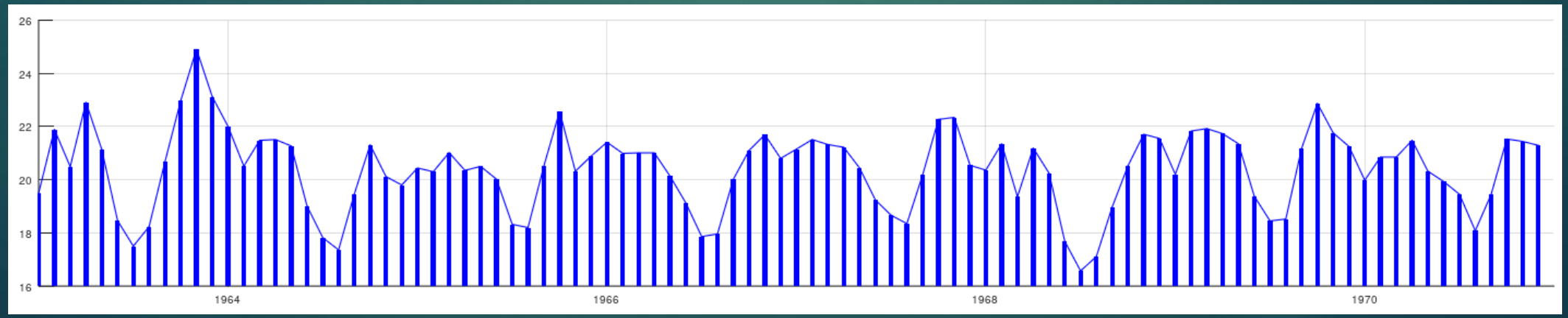
- São sistemas com realimentação, isto é, o valores da saída retornam para a entrada do sistema.
- A saída de sistemas IIR depende das entradas e de saídas passadas.
- A função de transferência de um sistema IIR tem tanto numerador quanto denominador

Série histórica das temperaturas do DF
média mensal desde 1963

```
>> [t,x]=textread("./Dados/temperatura_DF_2.txt","%f %f",686);
>> plot(t,x)
>> grid on;
>> axis([1962,2021,16,26])
```




```
>> hold on; grid on;
>> bar(t(1:96),x(1:96),0.3,'b','edgecolor','none')
>> plot(t(1:96),x(1:96),'b')
>> axis([1963,1971,16,26])
```



Exemplo – Calcular a média das últimas 12 amostras (média anual)

- Seja um sistema discreto H(z) que recebe na entrada uma sequência de valores x[n], e entrega na saída a sequencia y(n) , que é a média das doze últimas amostras de x[n].

$$x[\mathrm{n}] \xRightarrow{H(z)} y[\mathrm{n}] = \frac{1}{12}\sum_{k=0}^{11} x[\mathrm{n}-k]$$

$$y[\mathrm{n}] = \frac{1}{12}x[\mathrm{n}-11] + \frac{1}{12}x[\mathrm{n}-10] + \cdots + \frac{1}{12}x[\mathrm{n}-1] + \frac{1}{12}x[\mathrm{n}]$$

- 12 somas e 12 multiplicações
- Sistema direto, não-recursivo. A saída só depende das entradas, atual e passadas.

$$Y(z) = \frac{1}{12}z^{-11}X(z) + \frac{1}{12}z^{-10}X(z) + \cdots + \frac{1}{12}z^{-1}X(z) + \frac{1}{12}X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{12}z^{-11} + \frac{1}{12}z^{-10} + \cdots + \frac{1}{12}z^{-1} + \frac{1}{12}$$

$$h[\mathrm{n}] = \left[\frac{1}{12} \quad \frac{1}{12} \quad \frac{1}{12} \cdots \frac{1}{12}\right]$$

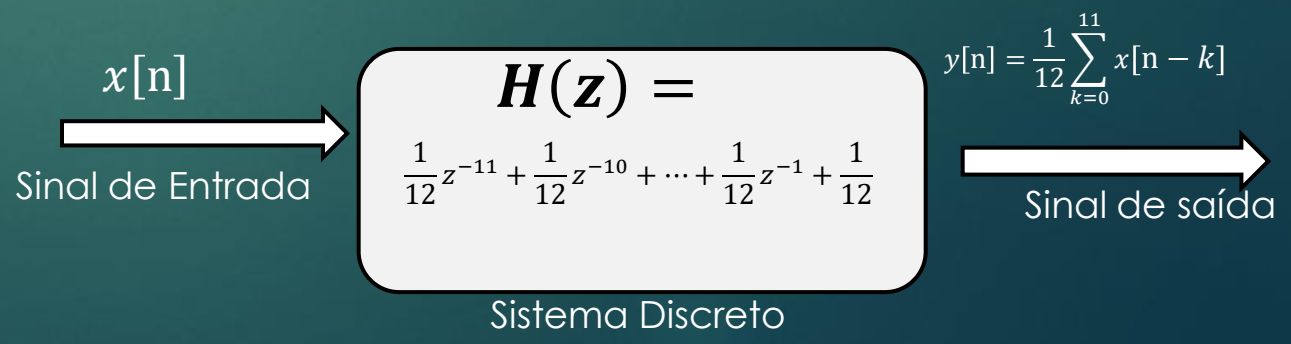
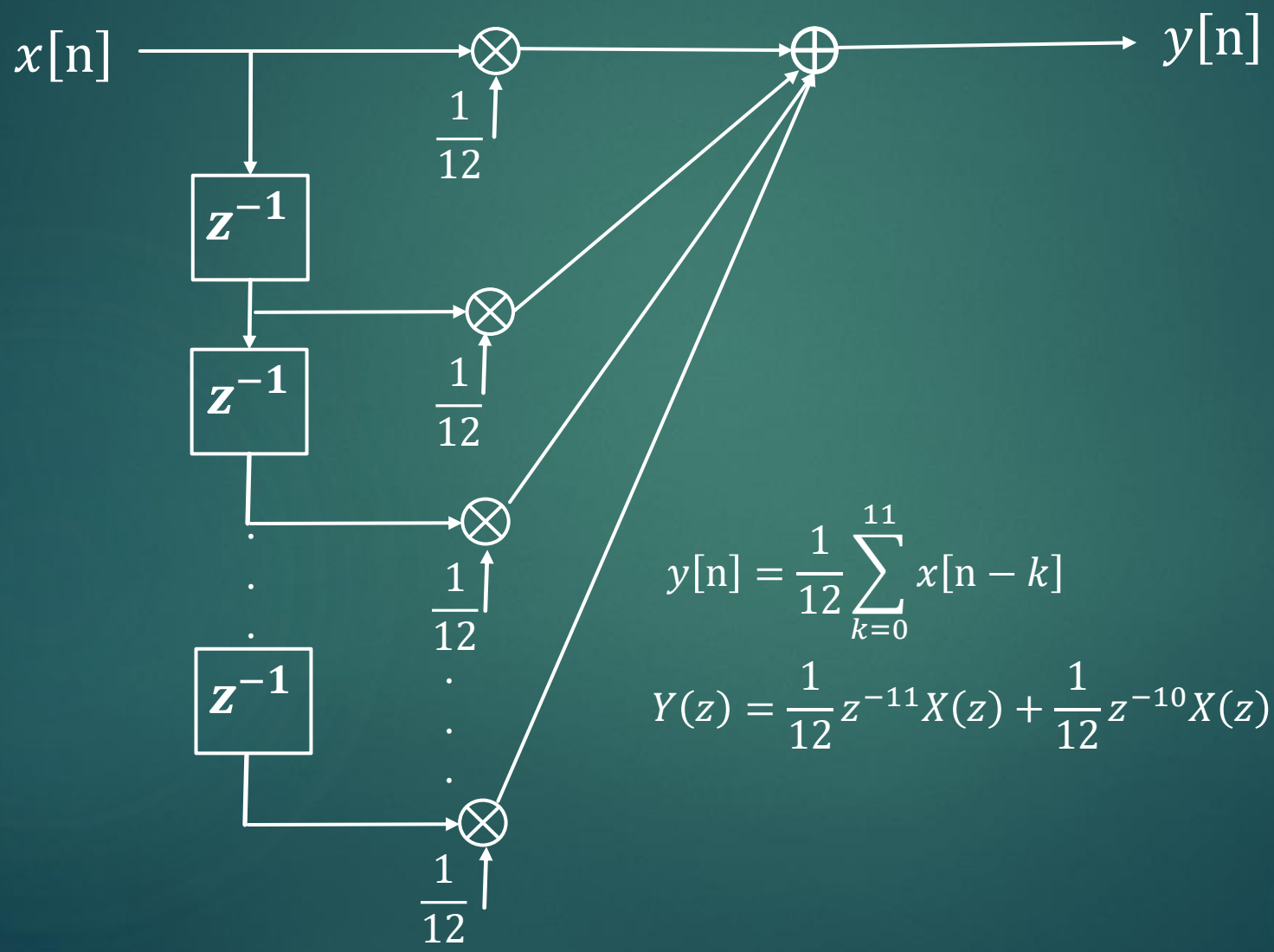


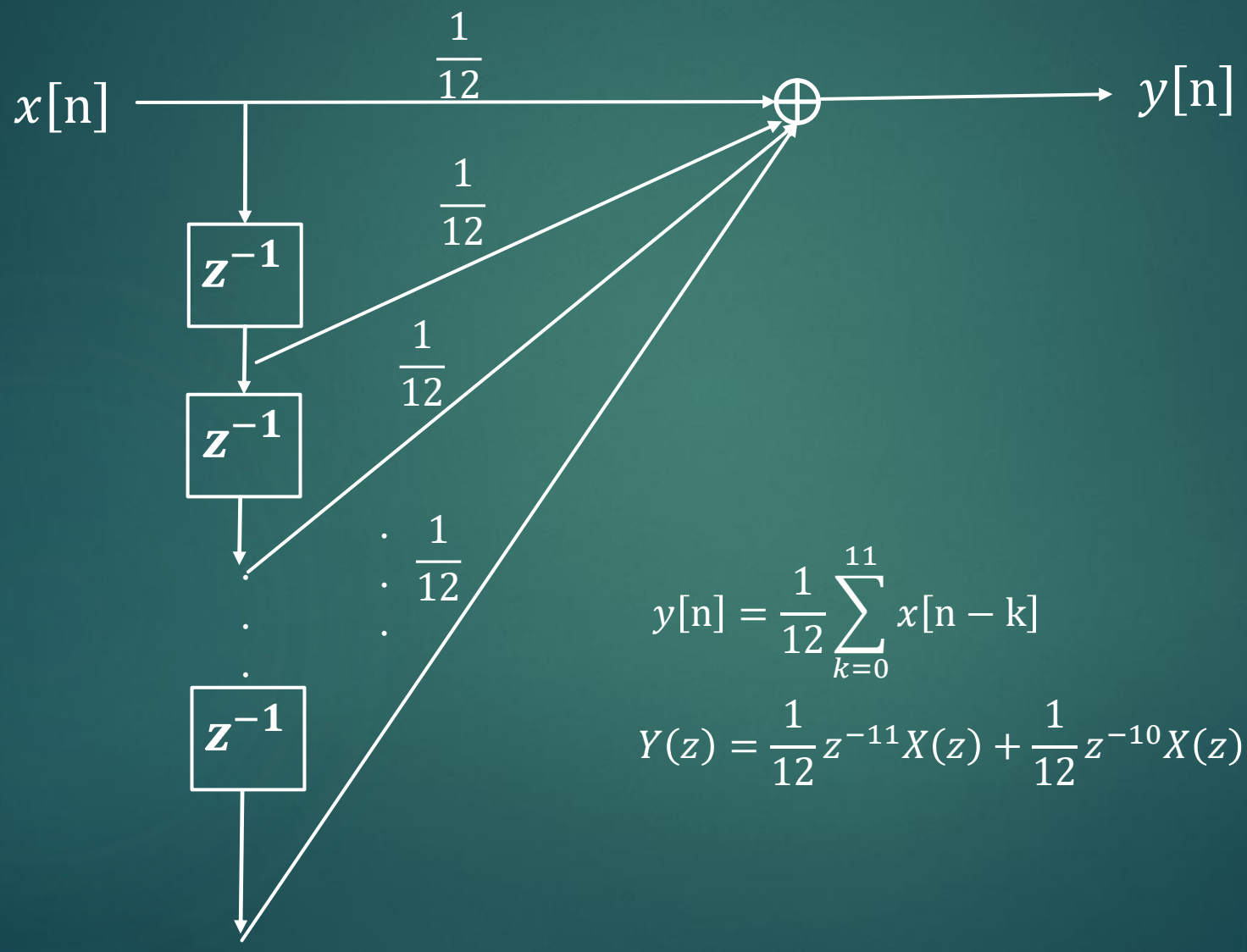
Diagrama de blocos da média FIR



$$y[\mathrm{n}] = \frac{1}{12}\sum_{k=0}^{11} x[\mathrm{n}-k]$$

$$Y(z) = \frac{1}{12}z^{-11}X(z) + \frac{1}{12}z^{-10}X(z) + \cdots + \frac{1}{12}z^{-1}X(z) + \frac{1}{12}X(z)$$

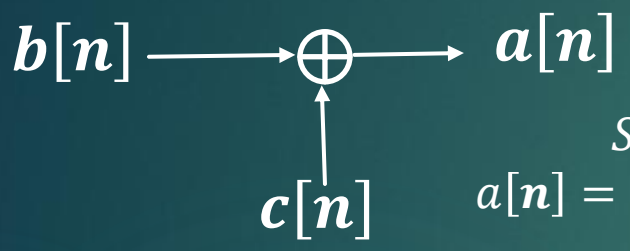
Diagrama de blocos média FIR



$$y[n] = \frac{1}{12}\sum_{k=0}^{11} x[n-k]$$

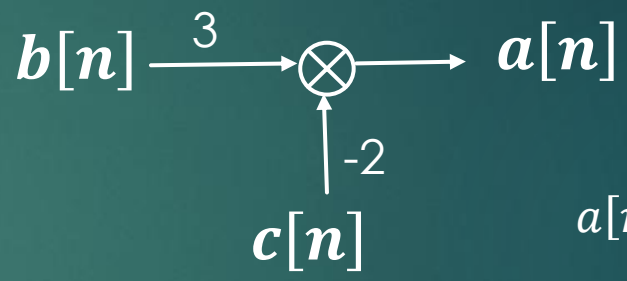
$$Y(z) = \frac{1}{12}z^{-11}X(z) + \frac{1}{12}z^{-10}X(z) + \cdots + \frac{1}{12}z^{-1}X(z) + \frac{1}{12}X(z)$$

Resumo Terminologia de Símbolos para Diagrama de Blocos



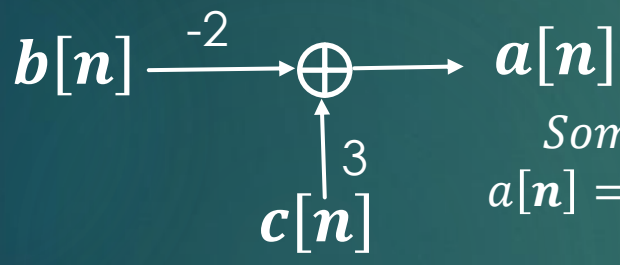
Soma

$a[n] = b[n] + c[n]$



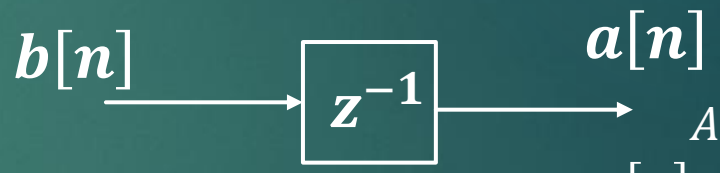
Multiplicação

$a[n] = 3b[n]\,(-2)c[n]$



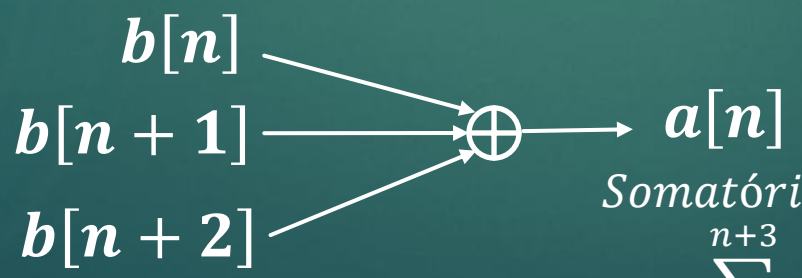
Soma Ponderada

$a[n] = -2b[n] + 3c[n]$



Atraso

$a[n] = b[n-1]$



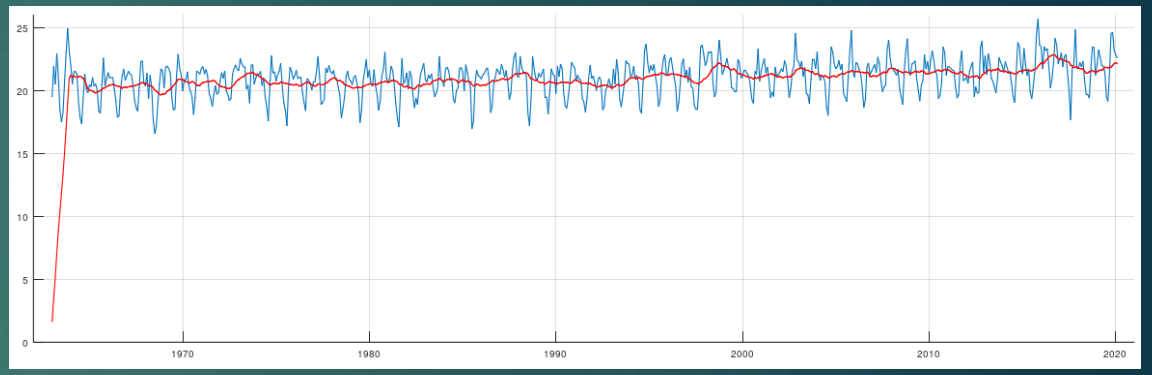
Somatória

$$a[n] = \sum_{k=n}^{n+3} b[n]$$

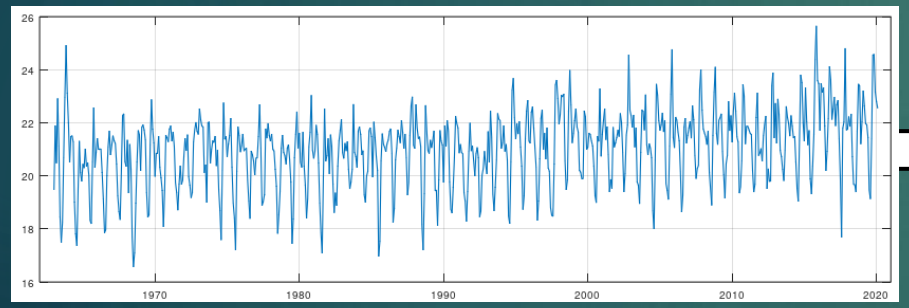
Resposta deste Sistema FIR à Entrada Temperatura

- Notar que só podemos calcular a média a partir do 12º mês, de modo que o valor da saída no 11 primeiros meses é indefinida.
- Ao tirarmos a média notamos uma sazonalidade e uma tendência da aumento da temperatura, que não víamos nos dados originais.

```
>> hold on; grid on;
>> y=rec3_FIR(x');
>> plot(t,x);
>> plot(t,y,'r');
```



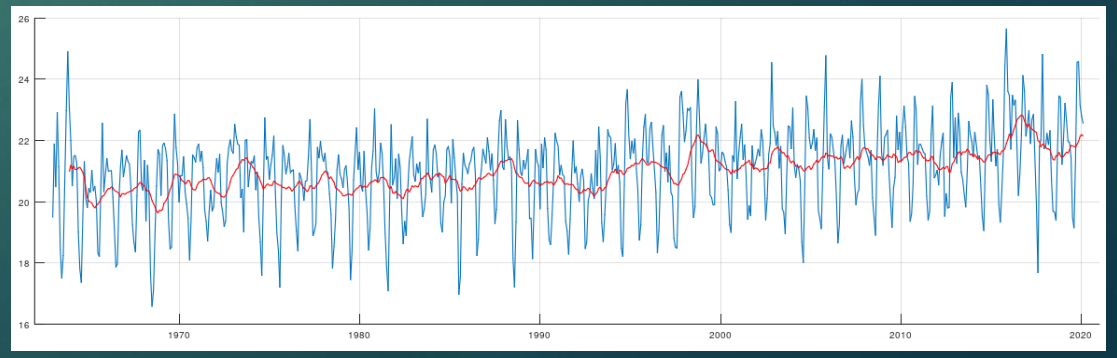

```
>> [t,x]=textread("./Dados/temperatura_DF_2.txt","%f %f",686);
>> plot(t,x)
>> grid on;
>> axis([1962,2021,16,26])
```




```
function y_out=rec3_FIR(x_in)
    xi=[zeros(1,11) x_in];
    for i=12:length(xi)
        yi(i)=mean( xi(i-11:i) );
    end
    y_out=yi(12:length(xi));
endfunction
```

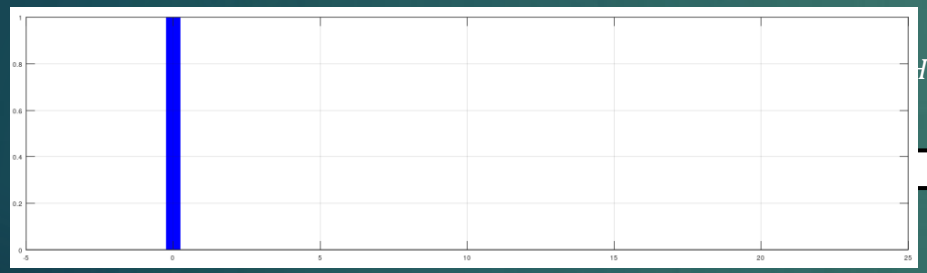
$$H(z) = \frac{1}{12}z^{-11} + \frac{1}{12}z^{-10} + \cdots + \frac{1}{12}z^{-1} + \frac{1}{12}$$

```
>> hold on; grid on;
>> y=rec3_FIR(x');
>> plot(t,x);
>> plot(t(12:end),y(12:end),'r');
```



Resposta deste Sistema FIR ao Impulso

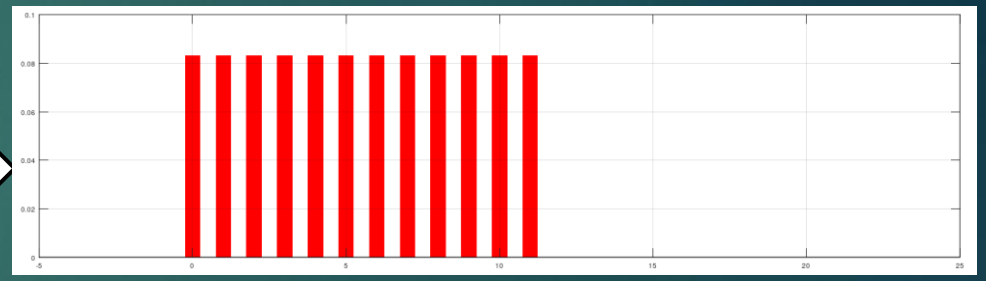
```
>> n=[0:20];
>> x=f_d(n);
>> bar(n,x,0.4,'b','edgecolor','none')
```



$$H(z) = \frac{1}{12}z^{-11} + \frac{1}{12}z^{-10} + \cdots + \frac{1}{12}z^{-1} + \frac{1}{12}$$

```
>> h=1/12*ones(1,12);
```

```
>> y=rec3_FIR(x);
>> bar(n,y,0.4,'r','edgecolor','none')
```

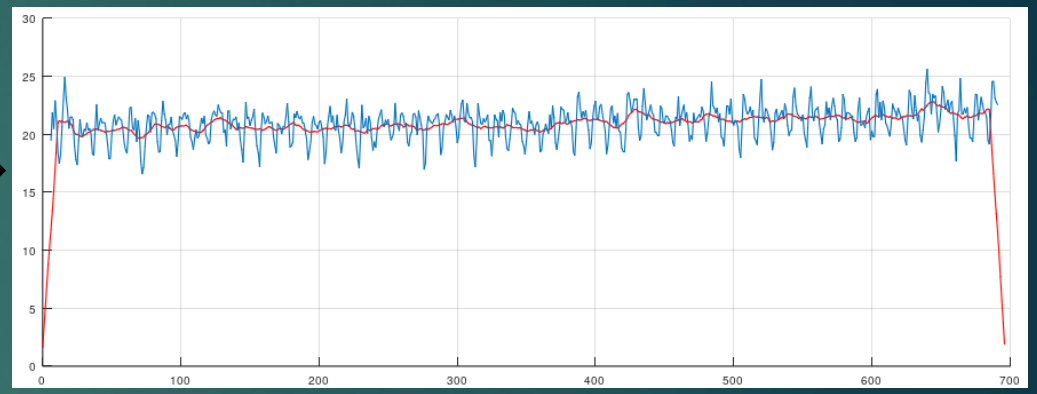


Então, a resposta ao impulso do sistema que faz a média das últimas 12 amostras é :

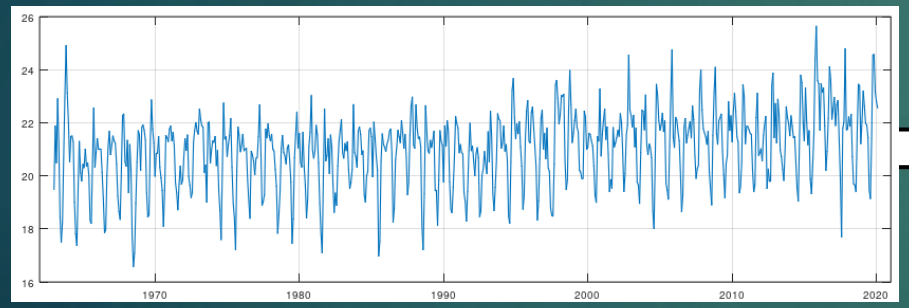
$$h[\boldsymbol{n}] = \left[\frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}\right]$$

Também podemos calcular a resposta deste Sistema FIR à entrada temperatura através da convolução da temperatura com a resposta ao impulso.

```
>> y=conv(x,h);
>> hold on; grid on;
>> plot([0:length(y)-1],y,'r')
>> plot([0:length(x)-1]+6,x);
```




```
>> [t,x]=textread("./Dados/temperatura_DF_2.txt","%f %f",686);
>> plot(t,x)
>> grid on;
>> axis([1962,2021,16,26])
```

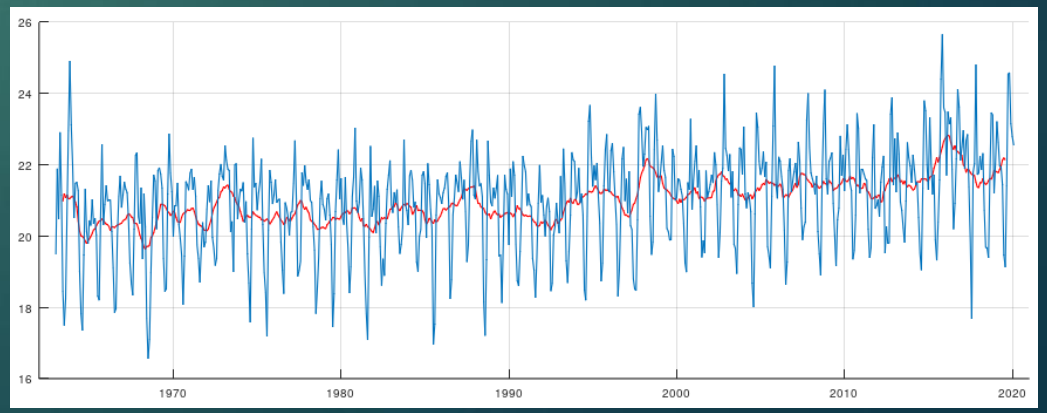


$$H(z) = \frac{1}{12}z^{-11} + \frac{1}{12}z^{-10} + \cdots + \frac{1}{12}z^{-1} + \frac{1}{12}$$

```
>> h=1/12*ones(1,12);
```

```
h=[1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 1/12 ];
```

```
>> hold on; grid on;
>> plot(t(6:Nx-6),y(12:Ny-11),'r')
>> plot(t,x)
```



Cálculo alternativo da média através de retroalimentação - Solução IIR

$$y[\boldsymbol{n}] = \frac{1}{12}\sum_{k=0}^{11} x[\boldsymbol{n}-\boldsymbol{k}]$$

$$y[n] = \frac{1}{12}x[n] + \frac{1}{12}\sum_{k=1}^{11} x\,[n-k] + \frac{1}{12}x[n-12] - \frac{1}{12}x[n-12]$$

O termo em amarelo foi retirado da somatória (índice k=0)

Os termos em laranja não alteram o valor de y(n), pois foi somado e subtraído mesmo valor

$$y[n] = \frac{1}{12}x[n] - \frac{1}{12}x[n-12] + \frac{1}{12}\sum_{k=1}^{11} x[n-12] + \frac{1}{12}x[n-12]$$

$$y[n-1] = \frac{1}{12}\sum_{k=1}^{11} x[n-k] + \frac{1}{12}x[n-12] = \frac{1}{12}\sum_{k=1}^{12} x[n-k]$$

Rearranjando a equação , vemos que o último termo em amarelo forma exatamente a saída anterior do sistema, isto é , $y[n-1]$

$$y[n] = \frac{1}{12}(x[n] - x[n-12]) + y[n-1]\ )$$

Cálculo alternativo da média com realimentação - Solução IIR

$$y[n] = \frac{1}{12}(\ x[n] - x[n-12]) \ + y[n-1]$$

$$x[n] \xRightarrow{H(z)} y[n] = \frac{1}{12}(\ x[n] - x[n-12]) + y[n-1]$$

Esta solução calcula a média com somente duas somas e uma multiplicação (e uma troca de sinal) !

No entanto o projeto é mais difícil e pode ser instável.

É um algoritmo recursivo, pois a saída atual depende das saídas passadas

$$Y(z) = \frac{1}{12}X(z) - \frac{1}{12}z^{-12}X(z) + \ z^{-1}Y(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$

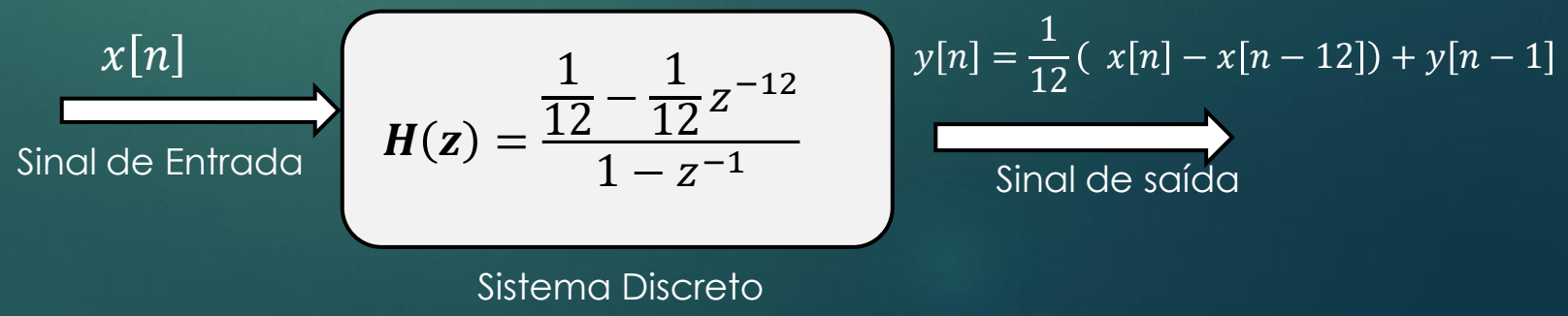
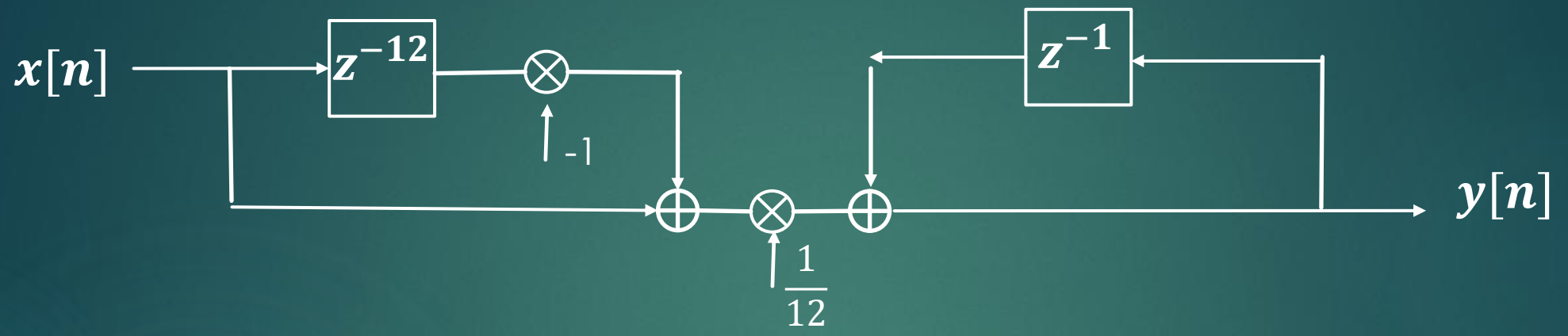
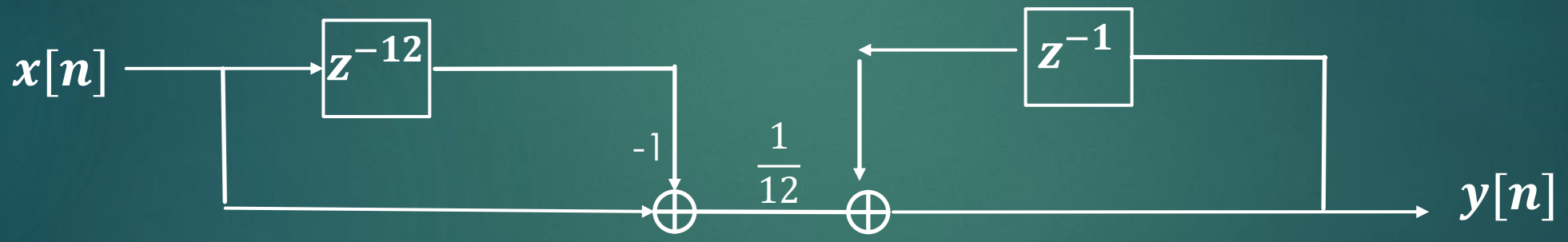


Diagrama de bloco média IIR



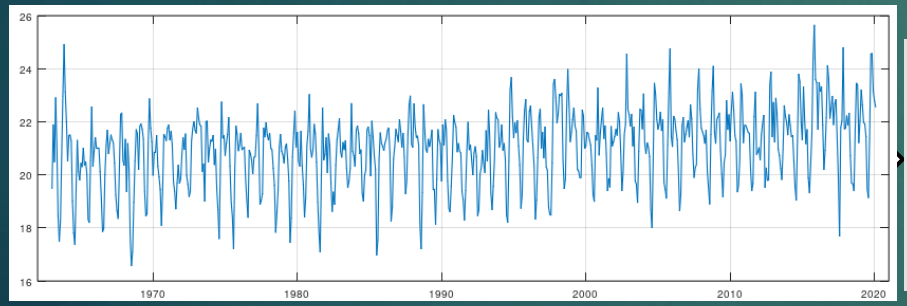


$$x[n] \xRightarrow{H(z)} y[n] = \frac{1}{12}(\ x[n] - x[n-12]) + y[n-1]$$

$$Y(z) = \frac{1}{12}X(z) - \frac{1}{12}z^{-12}X(x) + \ z^{-1}Y(z)$$

Resposta deste Sistema IIR à Entrada Temperatura

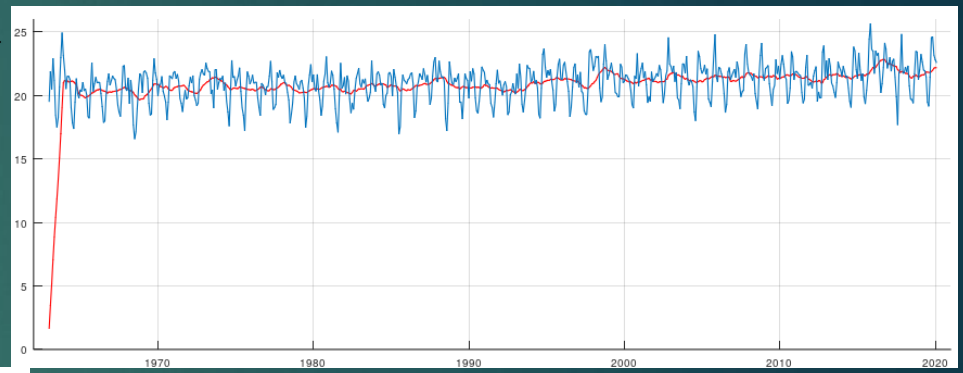
```
>> [t,x]=textread("./Dados/temperatura_DF_2.txt","%f %f",686);
>> plot(t,x)
>> grid on;
>> axis([1962,2021,16,26])
```



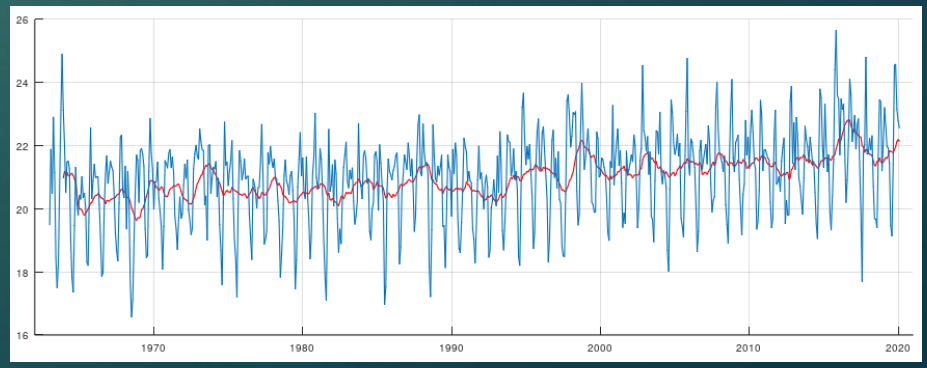
$$H(z) = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$

```
function y_out=rec3_IIR(x_in,y_ini)
    xi=[zeros(1,12) x_in];
    yi=[zeros(1,11)* nan  y_ini];
    for (i=13:length(xi))
        yi(i)=yi(i-1)+1/12*(xi(i)-xi(i-12));
    end
    y_out=yi(13:length(xi));
endfunction
```

```
>> y=rec3_IIR(x',0);
>> hold on; grid on;
>> plot(t,y,'r')
>> plot(t,x)
>> axis([1962,2021,0,26])
```




```
>> Nx=length(x);
>> hold on; grid on;
>> plot(t(12:N),y(12:N),'r');
>> plot(t,x)
>> axis([1962,2021,16,26])
```



- Notar que chegamos exatamente na mesma média.
- A escolha do valor inicial é fundamental para o sucesso de um filtro IIR

Resposta do Sistema à entrada degrau unitário
Condição inicial $y_{ini} = 0$

$$\boldsymbol{H(z)} = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$

$x[n]$

Sinal de Entrada

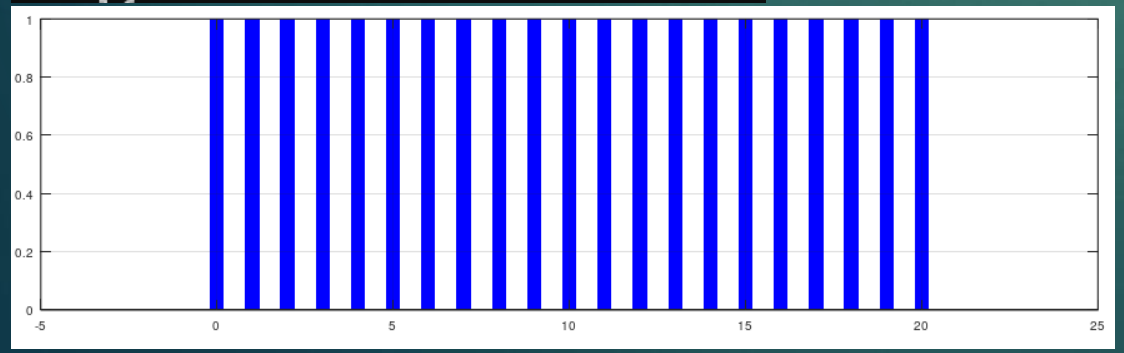
```
function y_out=rec3_IIR(x_in,y_ini)
    xi=[zeros(1,12) x_in];
    yi=[zeros(1,11)* nan  y_ini];
    for (i=13:length(xi))
        yi(i)=yi(i-1)+1/12*(xi(i)-xi(i-12));
    end
    y_out=yi(13:length(xi));
endfunction
```

Sistema Discreto

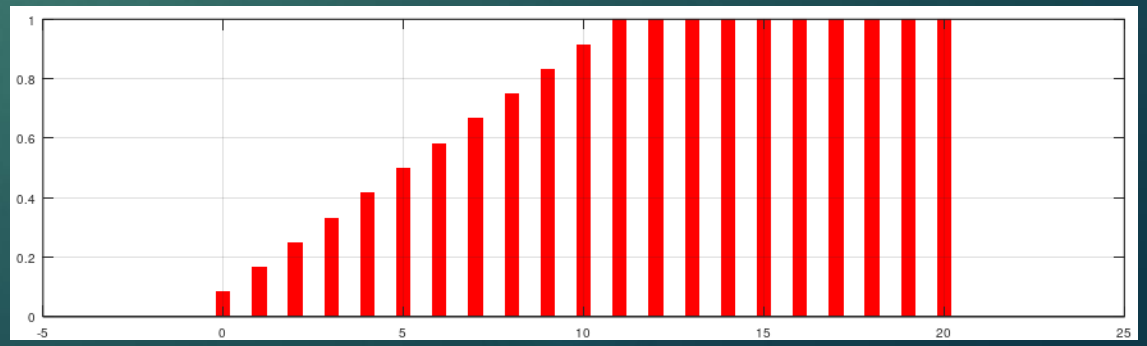
$$y[n] = \frac{1}{12}(\ x[n] - x[n-12]) + y[n-1]$$

Sinal de saída

```
>> n=[0:20];
>> x=f_u(n);
>> bar(n,x,0.4,'b','edgecolor','none')
```




```
>> y=rec3_IIR(x,0);
>> bar(n,y,0.4,'r','edgecolor','none')
```



Resposta do Sistema à entrada "cotação do dólar 2020"

$$\boldsymbol{H(z)} = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$

Sinal de Entrada

$x[n]$

```
function y_out=rec3_IIR(x_in,y_ini)
    xi=[zeros(1,12) x_in];
    yi=[zeros(1,11)* nan  y_ini];
    for (i=13:length(xi))
        yi(i)=yi(i-1)+1/12*(xi(i)-xi(i-12));
    end
    y_out=yi(13:length(xi));
endfunction
```

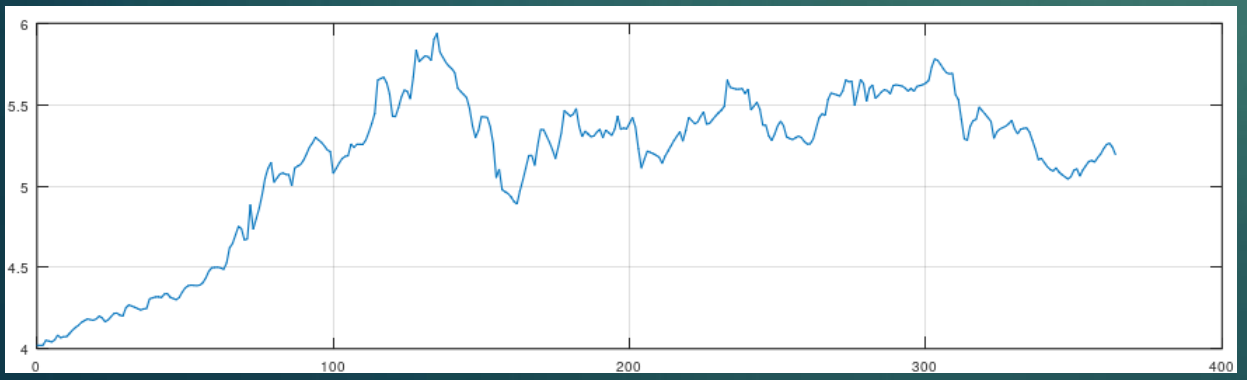
Sistema Discreto

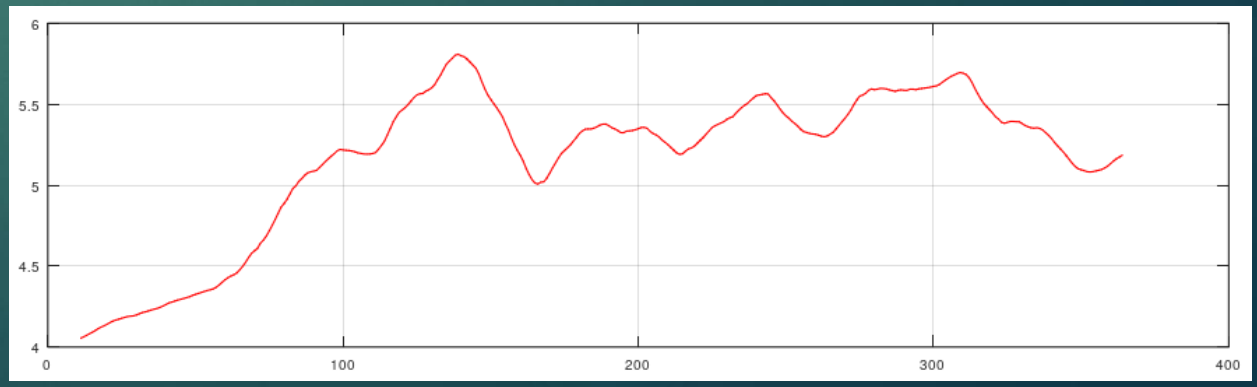
Sinal de saída

$$y[n] = \frac{1}{12}(\ x[n] - x[n-12]) + y[n-1]$$

```
>> A=load("./Dados/cotReal1.cvs");
>> t=A(:,1);
>> x=A(:,2);
>> plot(t,x)
```

```
>> y=rec3_IIR(x',0);
>> plot(t(12:end),y(12:end),'r')
```





Resposta do Sistema à entrada com 2 frequências + nível DC

$$\boldsymbol{H(z)} = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$

$x[n]$

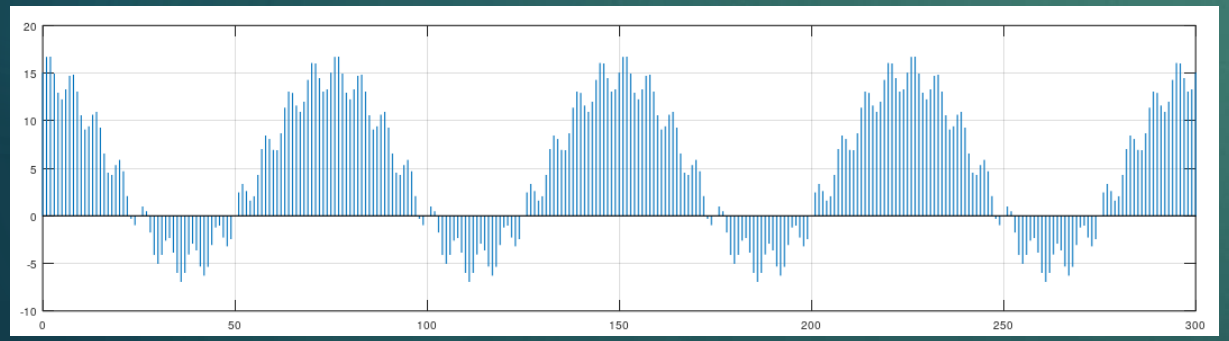
```
function y_out=rec3_IIR(x_in,y_ini)
    xi=[zeros(1,12) x_in];
    yi=[zeros(1,11)* nan  y_ini];
    for (i=13:length(xi))
        yi(i)=yi(i-1)+1/12*(xi(i)-xi(i-12));
    end
    y_out=yi(13:length(xi));
endfunction
```

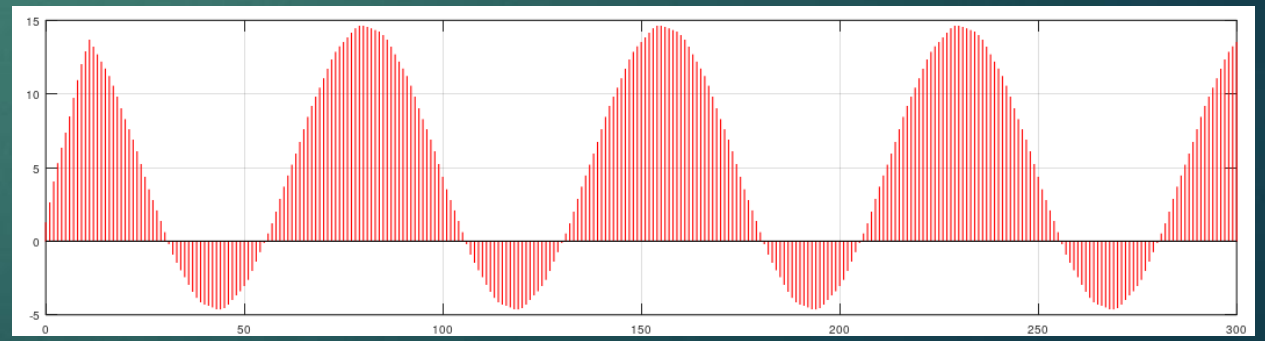
Sistema Discreto

$$y[n] = \frac{1}{12}(\ x[n] - x[n-12]) + y[n-1]$$

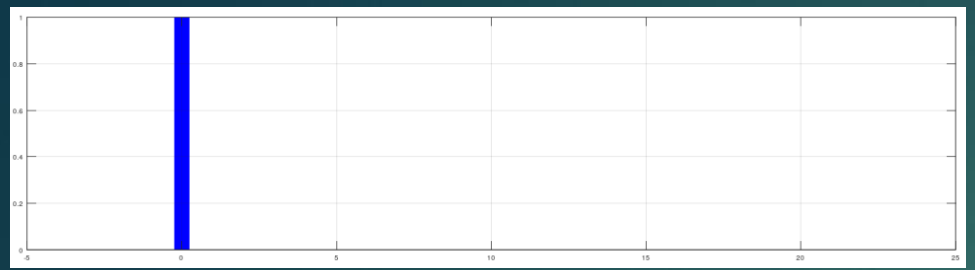
```
>> fx= @(t) (5+10*cos(8*pi/300*t)+2*sin(12*8*pi/300*t)).*f_u(t);
>> n=[0:300];
>> x=fx(n);
>> stem(n,x,'marker','none');
```

```
>> y=rec3_IIR(x,0);
>> stem(n,y,'r','marker','none');
```



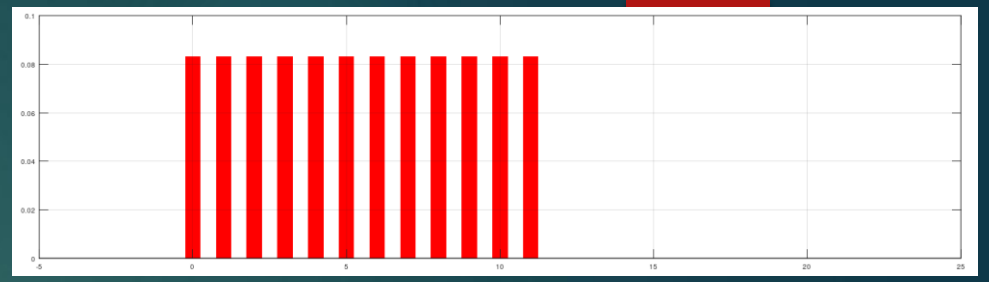


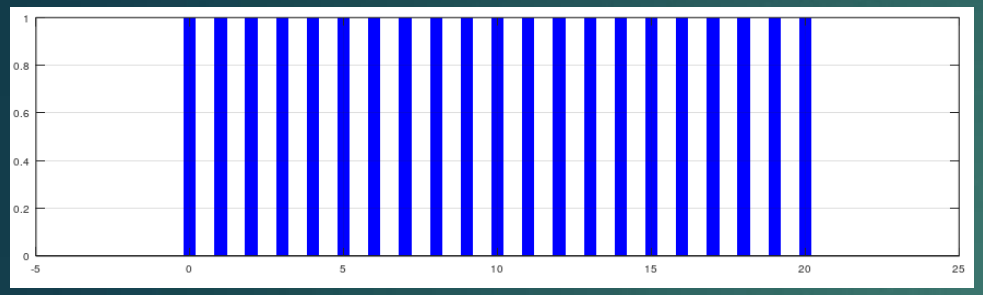
O Sistema removeu a frequência mais alta da entrada (12 rad/s) , preservando a frequência mais baixa (1 rad/s) e o nível DC



$x[n]$ → $$H(z) = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$ → $y[n]$

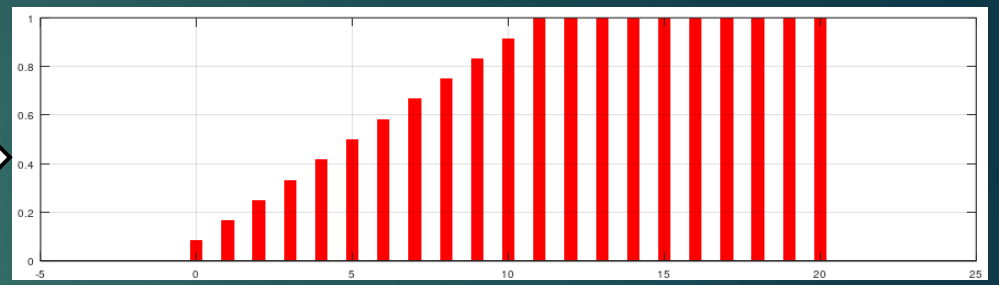
Impulso Unitário

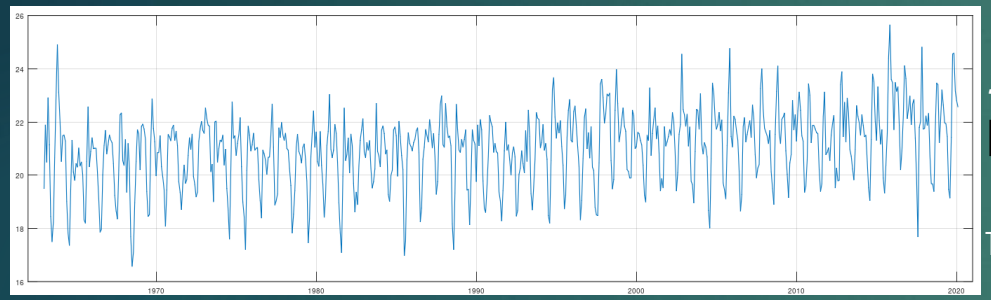




$x[n]$ → $$H(z) = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$ → $y[n]$

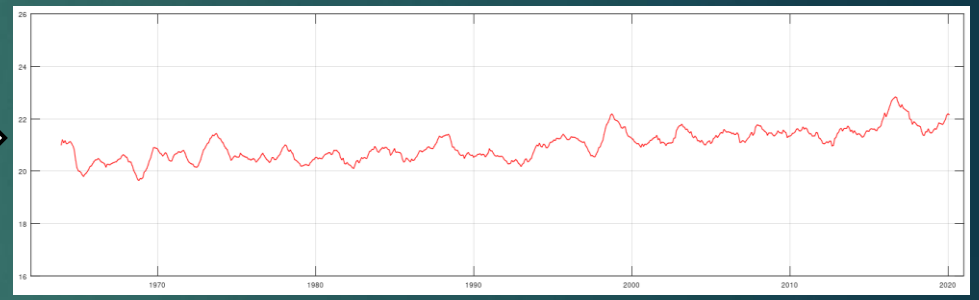
Degrau Unitário

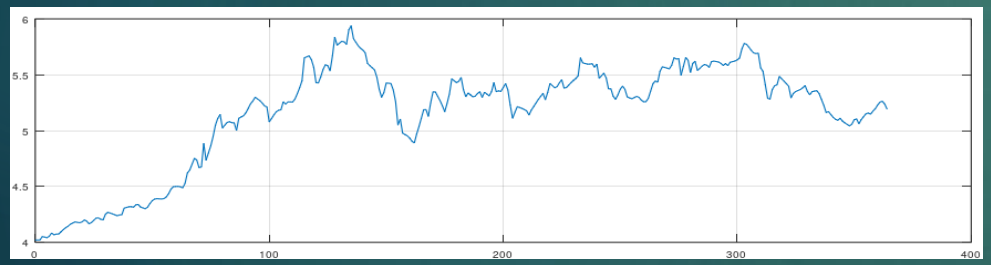




$x[n]$ → $$H(z) = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$ → $y[n]$

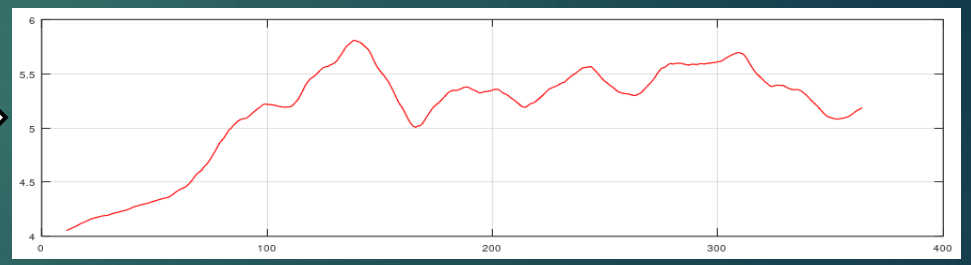
Temperatura mensal

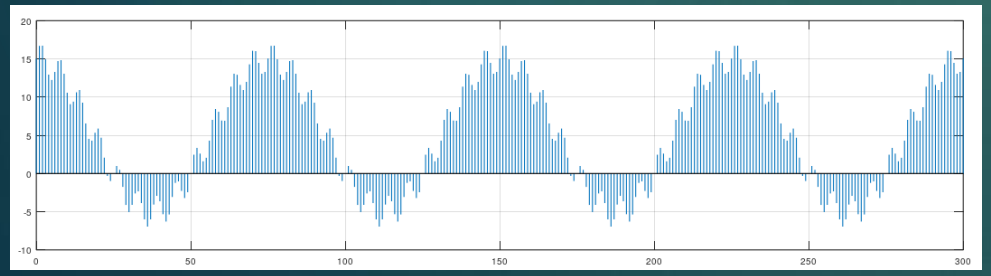




$x[n]$ → $$H(z) = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$ → $y[n]$

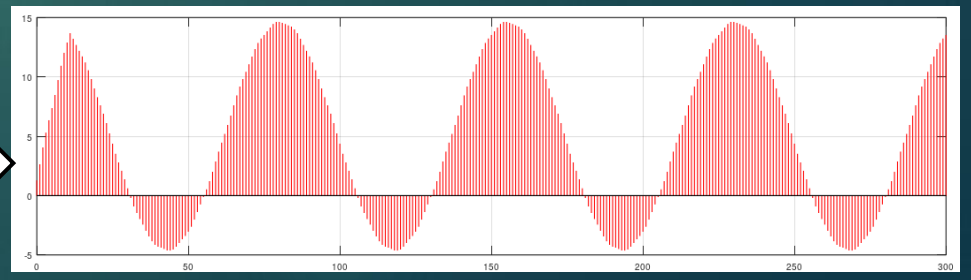
Dólar diário 2020





$x[n]$ → $$H(z) = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$ → $y[n]$

2 Frequência + DC



Testando a mesmas entradas com um outro Sistema LP

Passa Baixas, que retira as altas frequências (variações rápidas) e preserva as tendências

$x[n]$

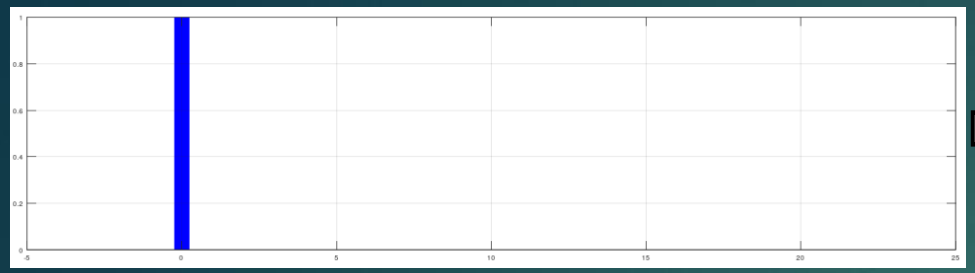
Sinal de Entrada

```
function y_out=Sistema_LP(x_in,y1,y2,y3)
    xi=[0 0 0  x_in];
    yi=[y3 y2 y1];
    for (i=4:length(xi))
       yi(i)=0.8771*yi(i-1)-0.2455*yi(i-2)+0.02224*yi(i-3)+...
       0.04328*xi(i)+0.1298*xi(i-1)+0.1298*xi(i-2)+0.04328*xi(i-3);
    end
    y_out=yi(4:length(xi));
endfunction
```

$y[n]$

$$H(z) = \frac{0.04328 + 0.1298z^{-1} + 0.1298z^{-2} + 0.04328z^{-3}}{1 - 0.8771z^{-1} + 0.2455z^{-2} - 0.02224z^{-3}}$$

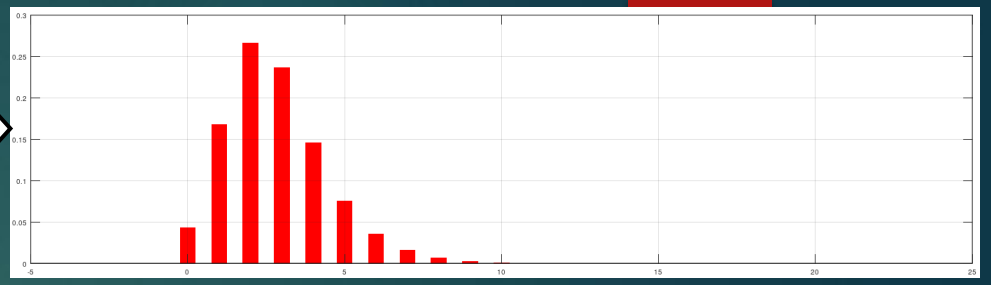
$$y[n]=0.04328\,x[n] + 0.1298x[n-1] + 0.1298x[n-2] + 0.04328x[n-3]$$
$$+0.8771\,y[n-1] - 0.2455\,y[n-2] + 0.02224\,y[n-3]$$



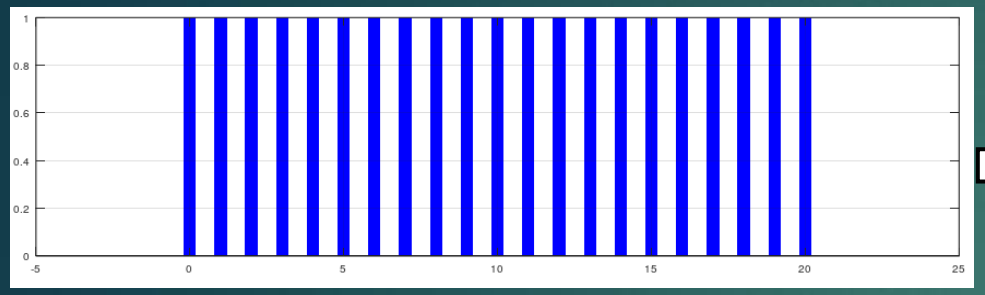
$x[n]$

$$\boldsymbol{H(z)} = \frac{0.04 + 0.13z^{-1} + 0.13z^{-2} + 0.04z^{-3}}{1 - 0.88z^{-1} + 0.25z^{-2} - 0.02z^{-3}}$$

$y[n]$



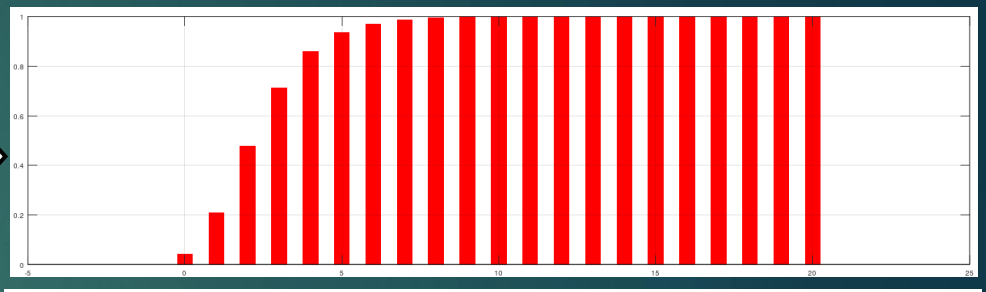
Temperatura mensal



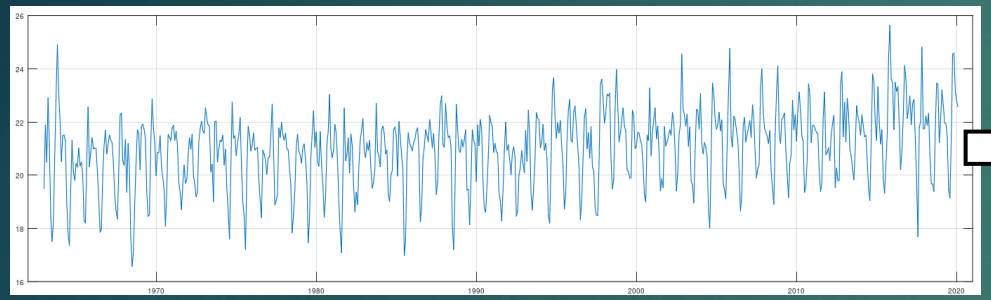
$x[n]$

$$\boldsymbol{H(z)} = \frac{0.04 + 0.13z^{-1} + 0.13z^{-2} + 0.04z^{-3}}{1 - 0.88z^{-1} + 0.25z^{-2} - 0.02z^{-3}}$$

$y[n]$



Impulso Unitário



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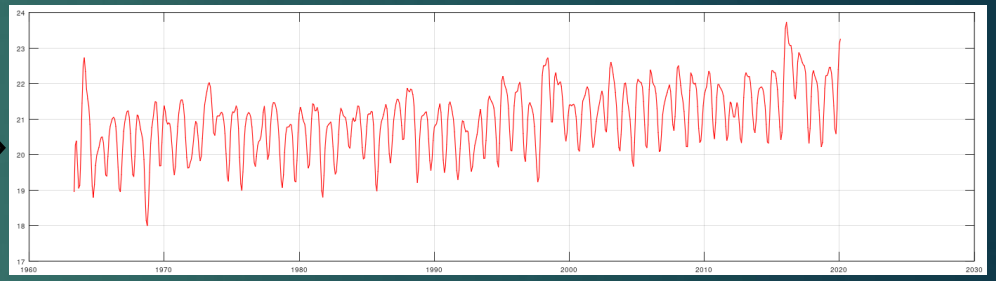
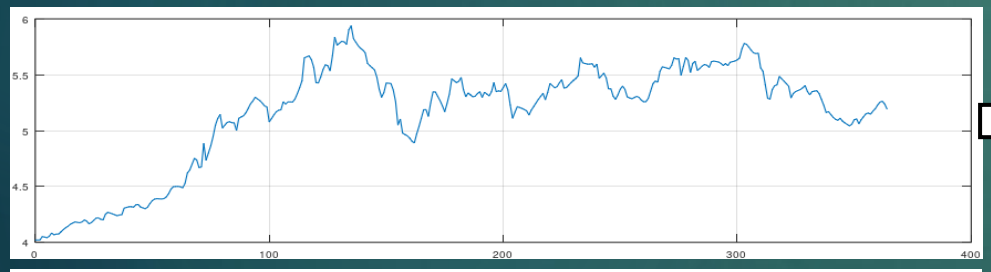


Degrau Unitário



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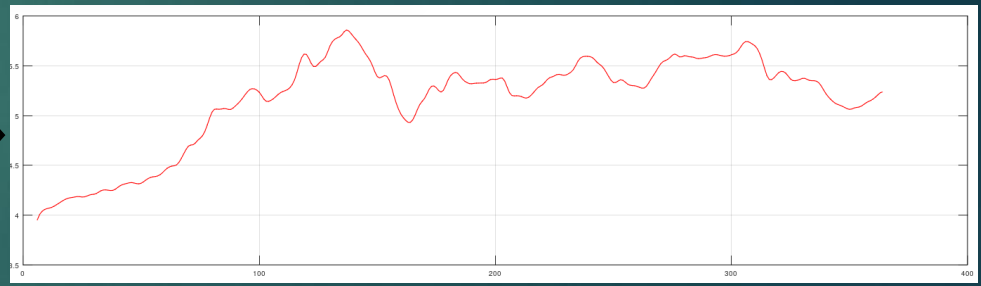
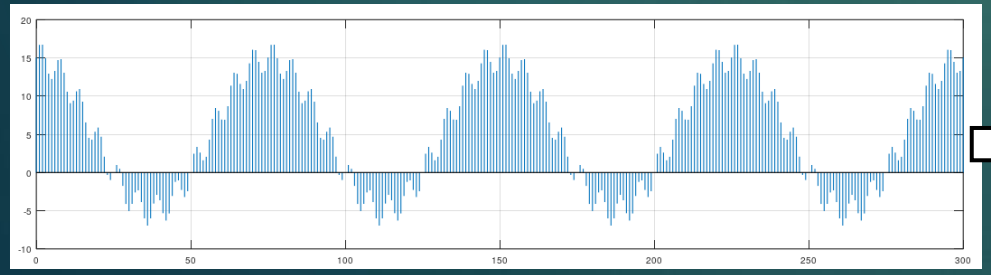
$y[n]$



Dólar diário 2020



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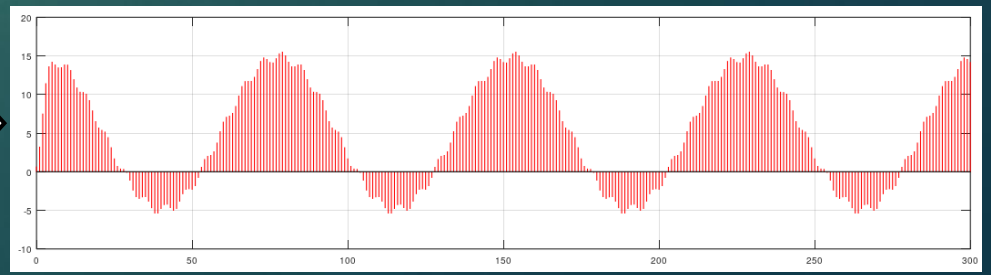
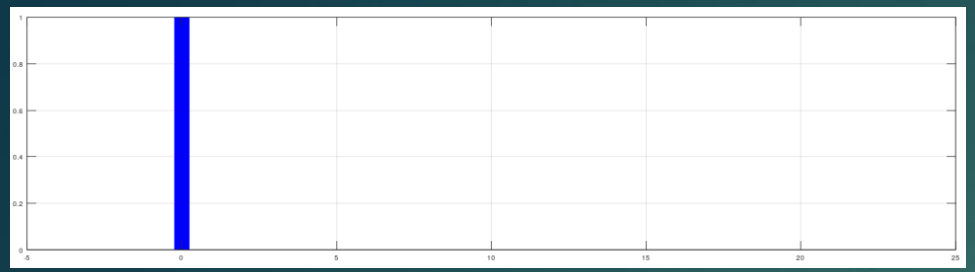
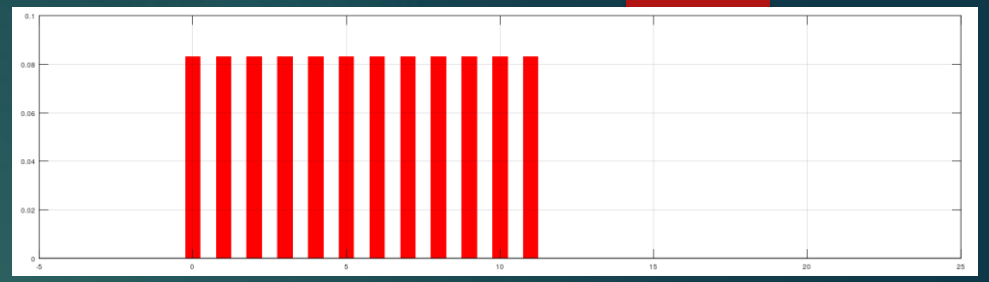


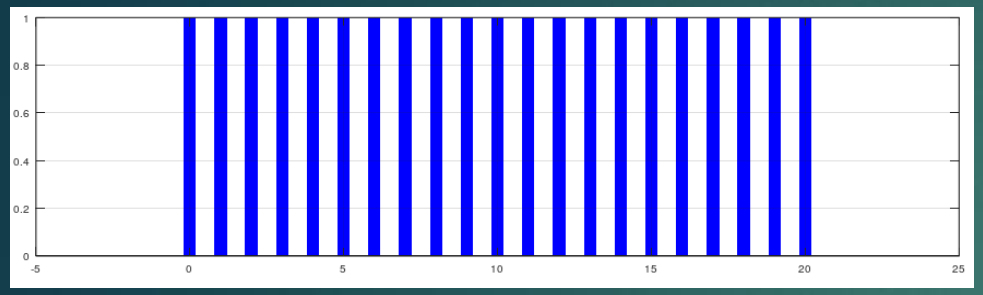
2 Frequência + DC



$x[n]$ → $$H(z) = \frac{\frac{1}{12} - \frac{1}{12}z^{-12}}{1 - z^{-1}}$$ → $y[n]$



Impulso Unitário



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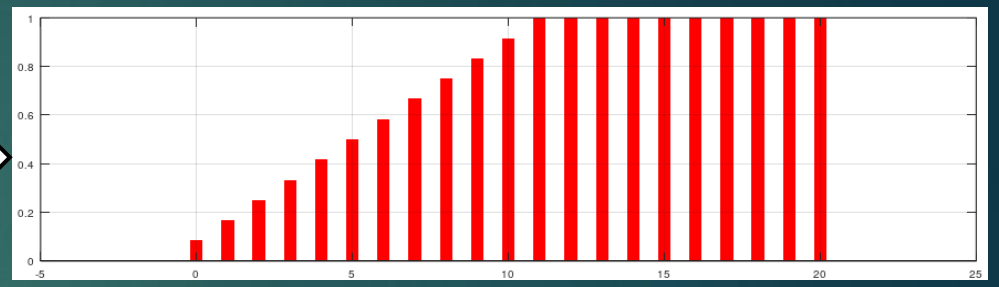
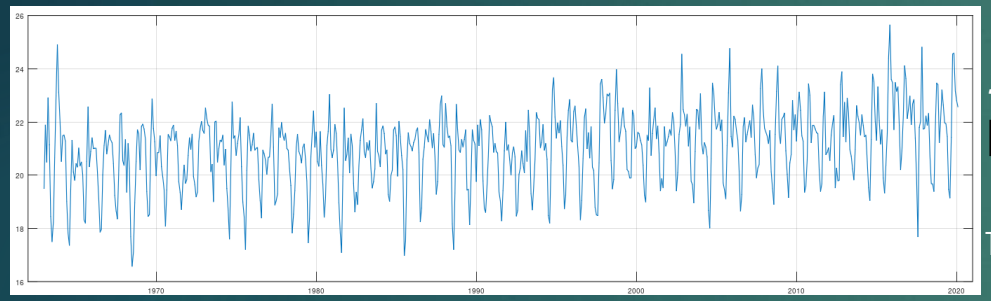


Degrau Unitário



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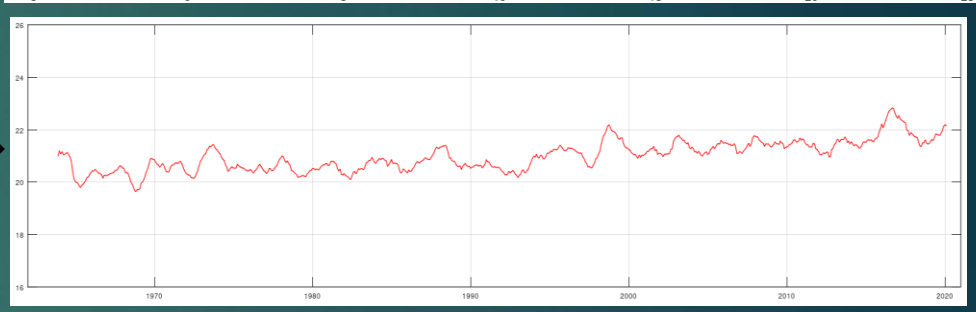
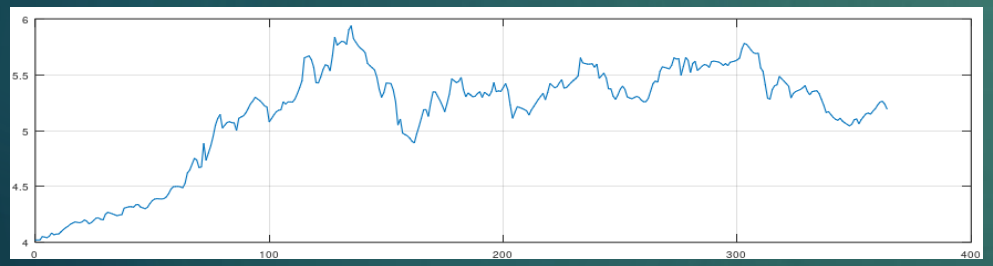


Temperatura mensal



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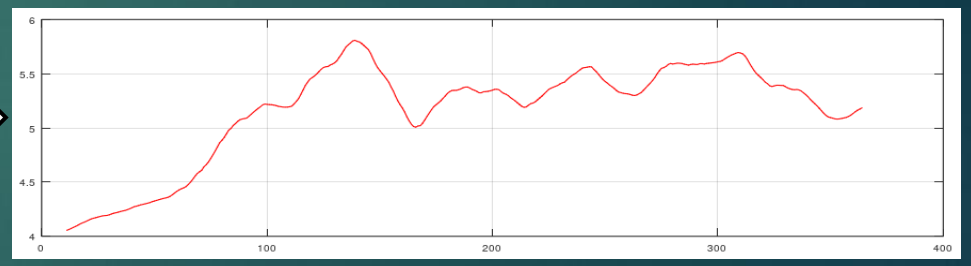
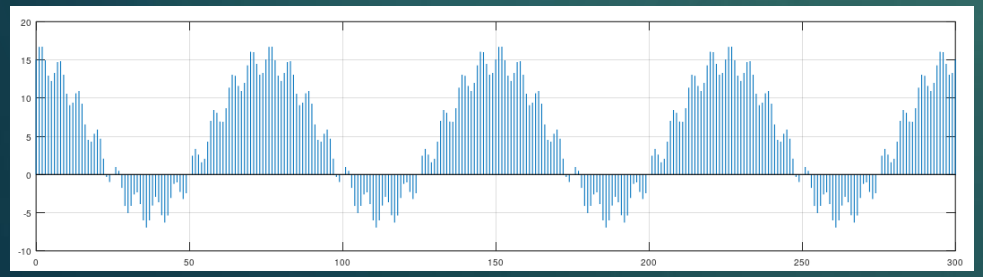


Dólar diário 2020



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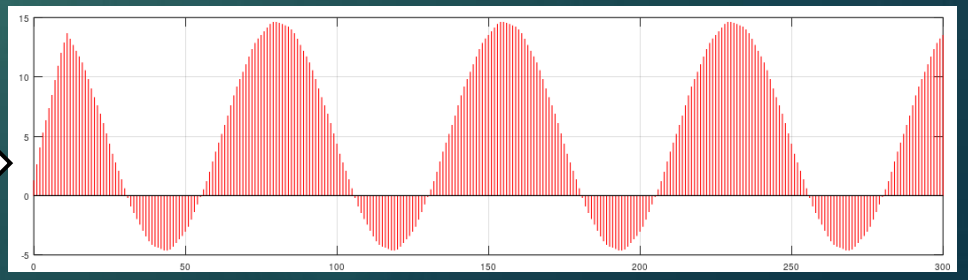


2 Frequência + DC

Testando a mesmas entradas com um outro Sistema HP

Passa Altas, retiras as baixas frequências (tendência), preservando as variações rápidas

$$y[n] = x[n] + 2x[n-1] + 4x[n-2] - 4x[n-3] - 2x[n-4] - x[n-5]$$

$x(n)$

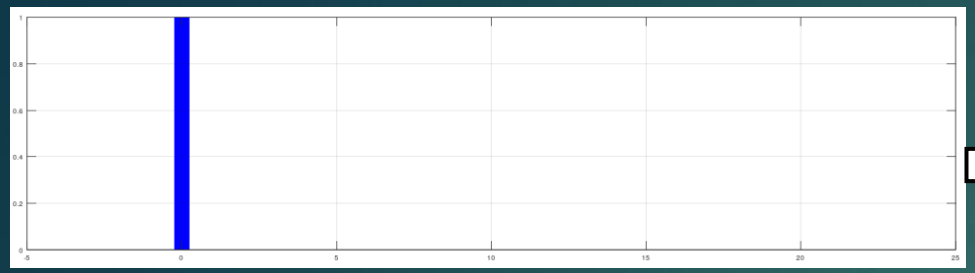
Sinal de Entrada

```
function y_out=Sistema_HP(x_in)
    xi=[zeros(1,5) x_in];
    for (i=6:length(xi))
        yi(i)=   xi(i)  +2*xi(i-1) +4*xi(i-2)...
            -4*xi(i-3)-2*xi(i-4) -xi(i-5);
    end
    y_out=yi(6:length(xi));
endfunction
```

Sinal de saída

$$H(z) = 1 + 2z^{-1} + 4z^{-2} - 4z^{-3} - 2z^{-4} - z^{-5}$$

$$y[n] = x[n] + 2x[n-1] + 4x[n-2] - 4x[n-3] - 2x[n-4] - x[n-5]$$

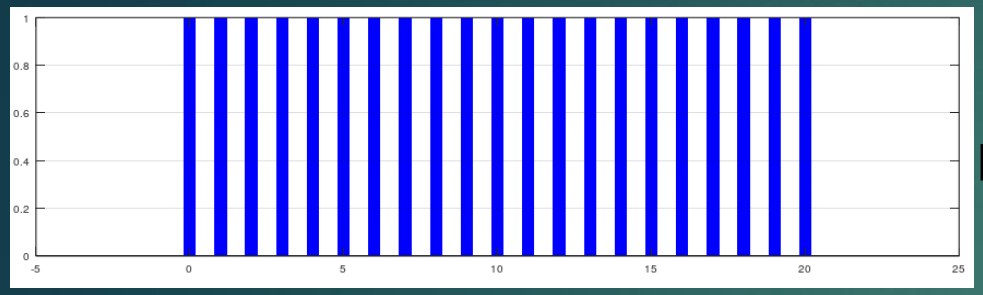


$x[n]$

$$\boldsymbol{H(z)} = 1 + 2z^{-1} + 4z^{-2} - 4z^{-3} - 2z^{-4} - z^{-5}$$

Temperatura mensal

$y[n]$

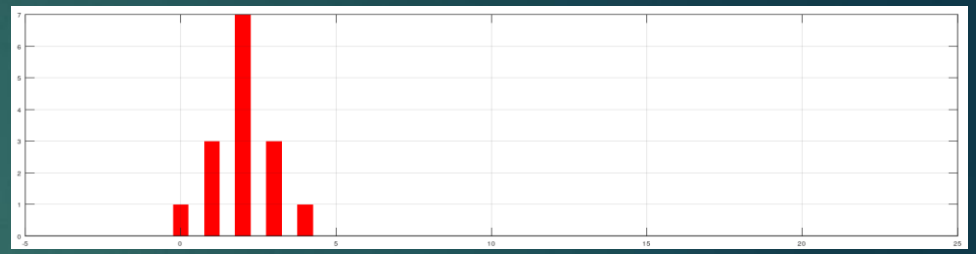


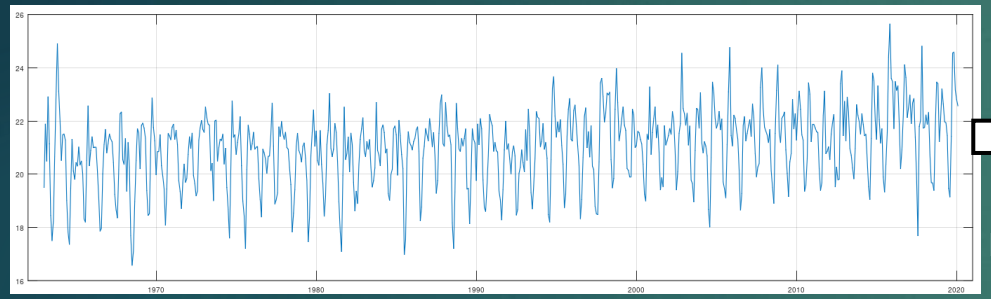
$x[n]$

$$\boldsymbol{H(z)} = 1 + 2z^{-1} + 4z^{-2} - 4z^{-3} - 2z^{-4} - z^{-5}$$

Impulso Unitário

$y[n]$





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Degrau Unitário

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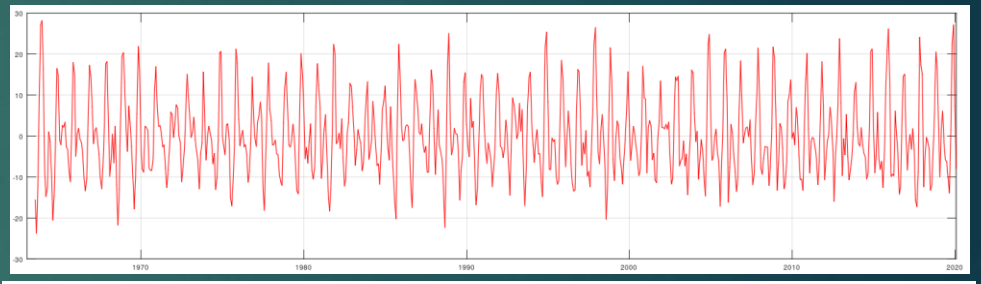

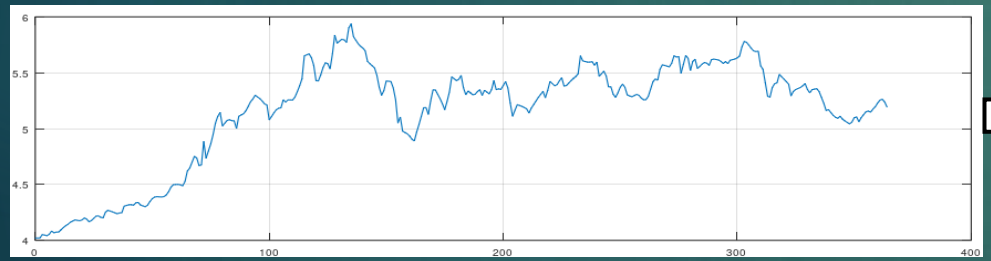

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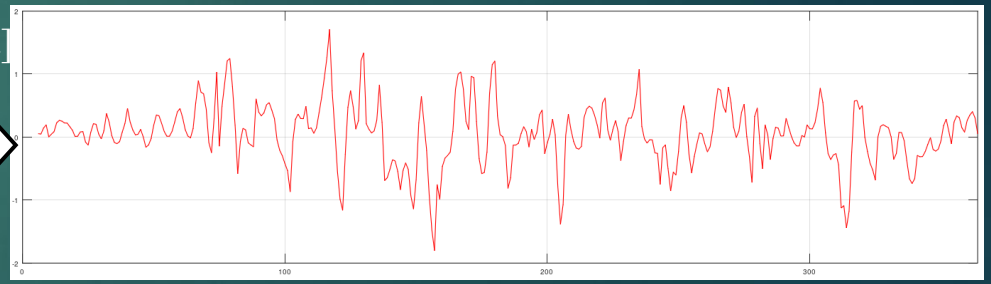
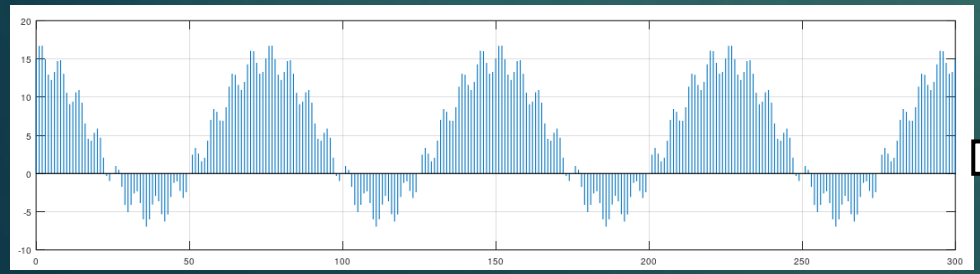
Dólar diário 2020

$y[n]$





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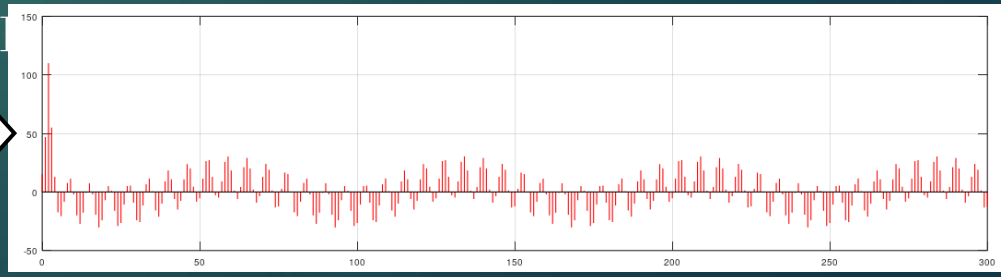
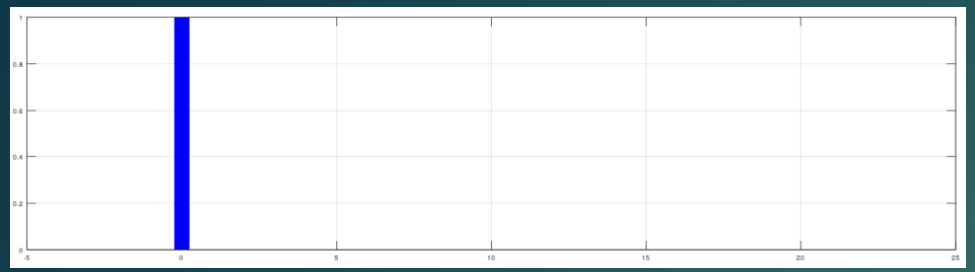
2 Frequência + DC

$y[n]$





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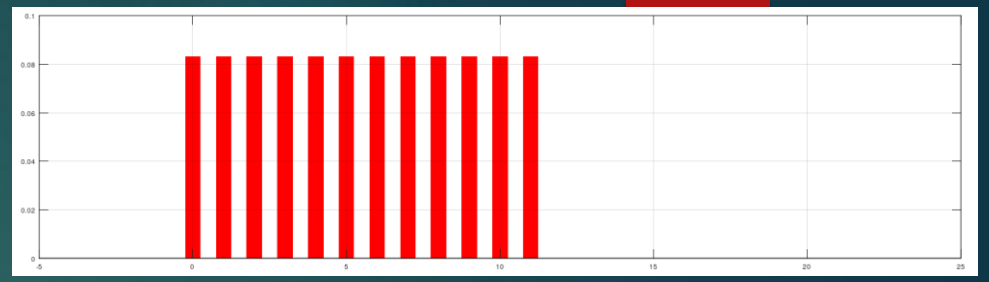
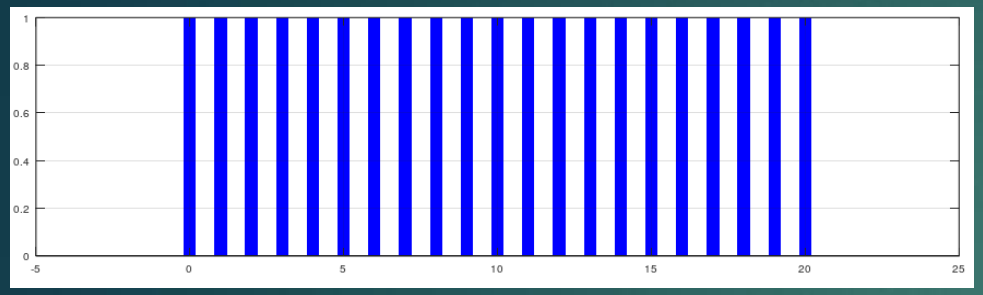


Impulso Unitário



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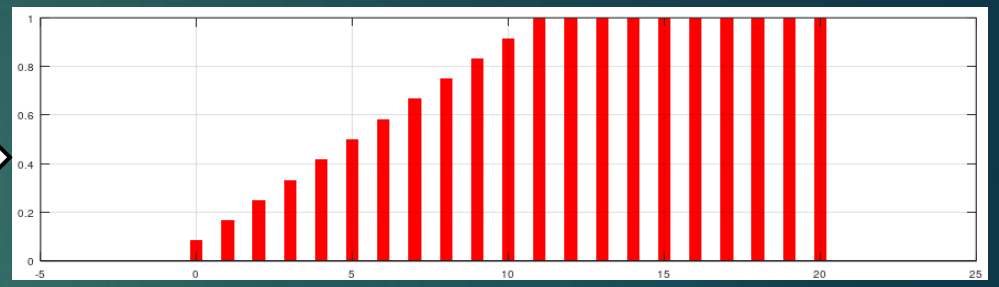
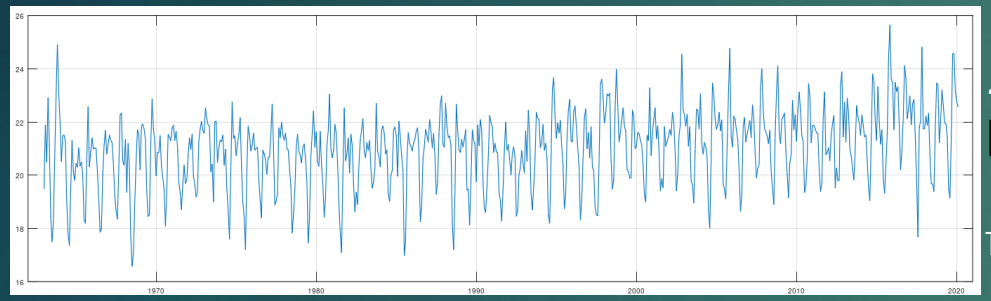


Degrau Unitário



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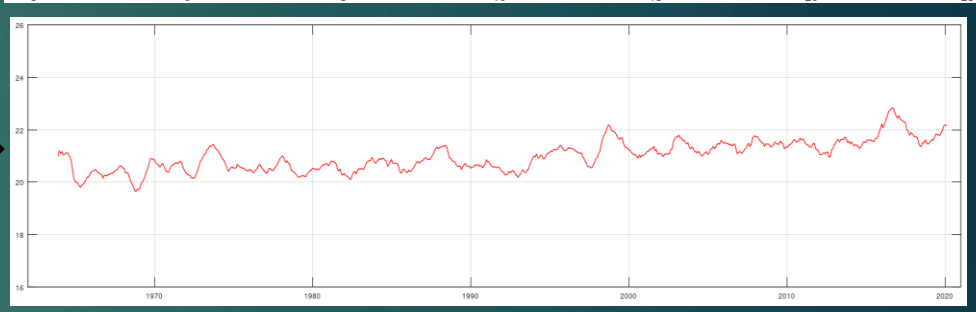
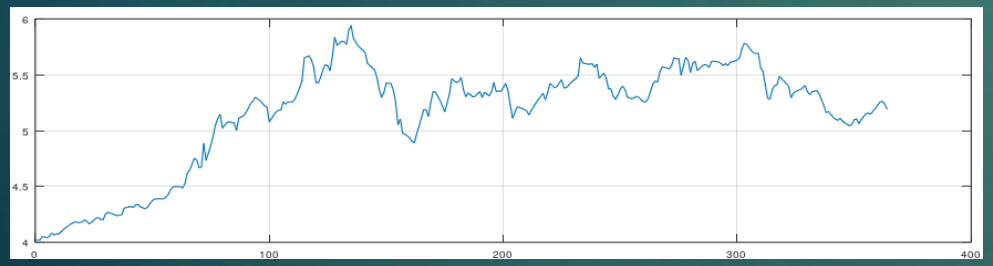


Temperatura mensal



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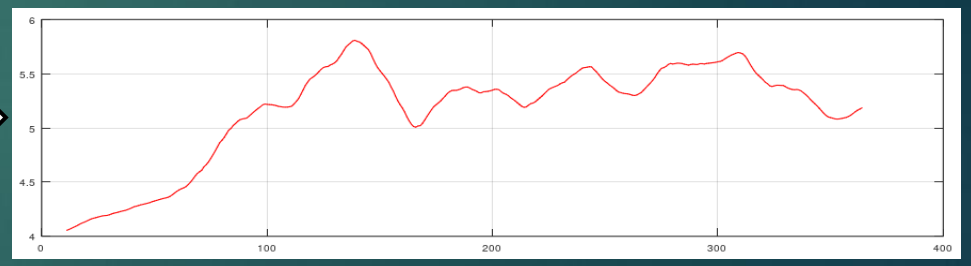
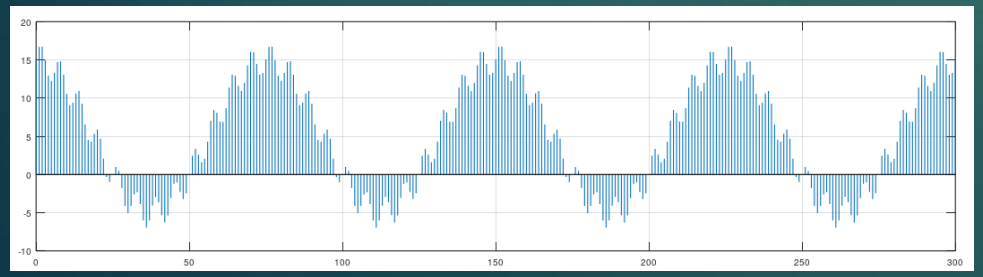


Dólar diário 2020



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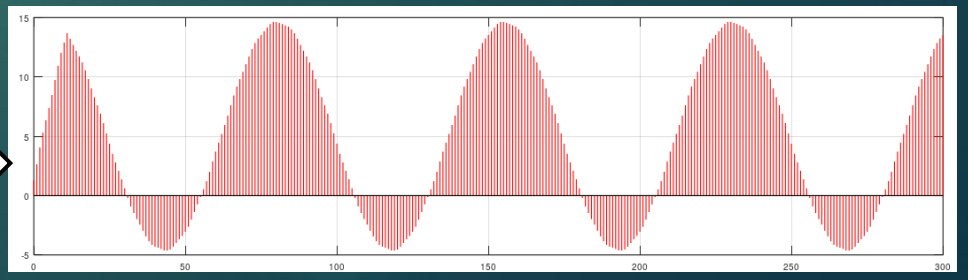

2 Frequência + DC